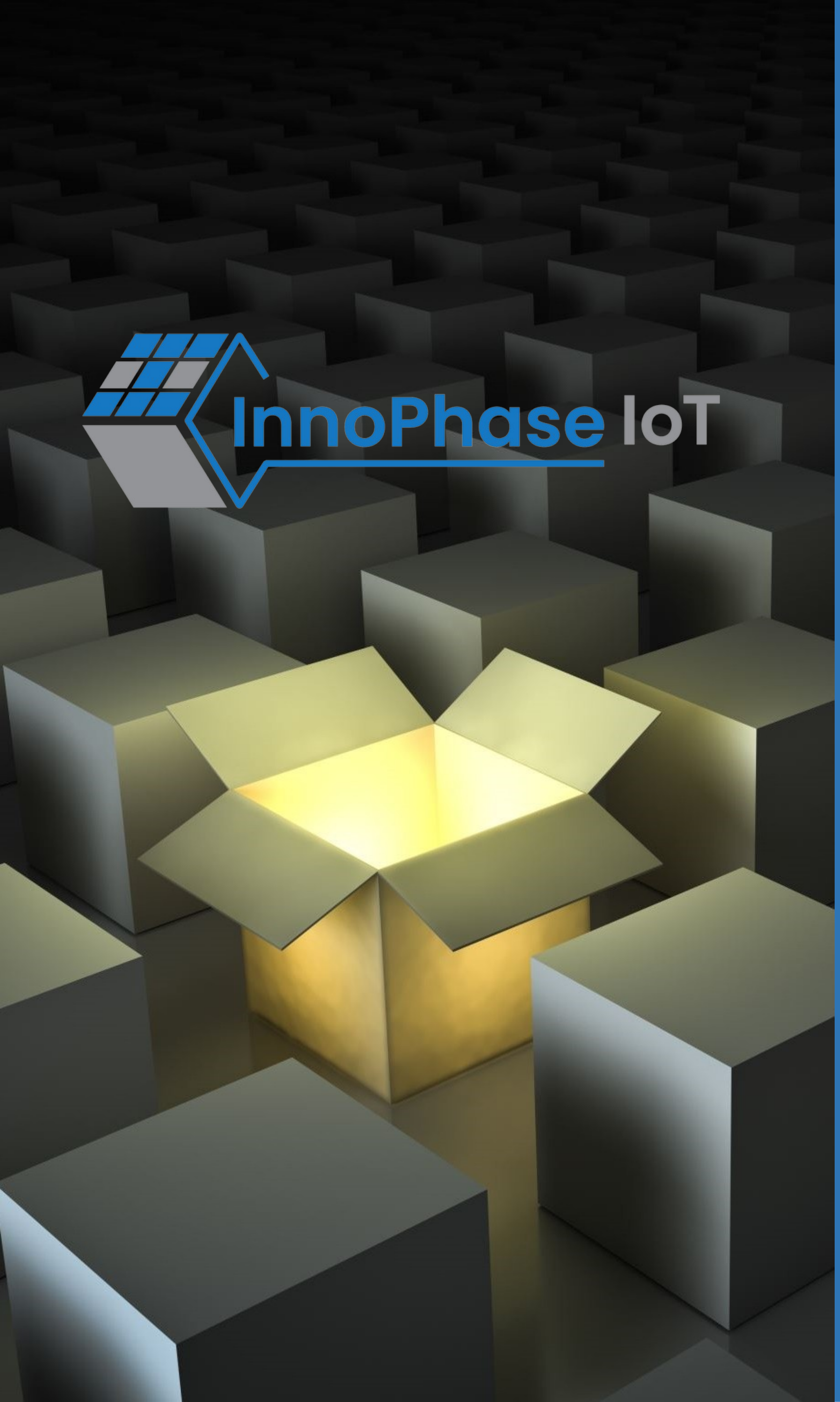


USDK1.1.1-Rel7 – Neon.B0 SW Release Readiness

4/10/2026

USDK 1.1.1-Rel7 Neon.B0 Location

SDK Package is located at neon_b0_USDK1.1.1-202615_rc7.3.zip

- docs
- prebuilt
- source
- tools
- neon.b0-apu-changelog.pdf
- neon.b0-npu-changelog.pdf
- neon.b0-release-notes.pdf

Note: Please use this link to navigate to the SDK above if download takes time Neon-B0

USDK 1.1.1-Rel7 Neon.B0

Change List for Release 7

Highlights

1. **Wi-Fi:** Added Wi-Fi 7 support
2. **Stability:** Overall stability improved
3. **Performance:** Improved 11ax performance
4. **Peripherals:** Added PDL driver support for Comparator, I2S, GPIO, SPI, and WDT
5. **BLE:** Significantly improved 1M PHY throughput
6. **Application Protocols:** Added support for MQTT, HTTP, and SNTP
7. **Bug Fixes:** Resolved issues in Wi-Fi and BLE

Lowlights

8. **Wi-Fi:** Performance variations with BE88U AP under investigation
9. **Interoperability:** Intermittent data stalls observed on select APs
10. **Hardware:** Board variation observed on both DTIM and system performance testing

Peripheral Support Status

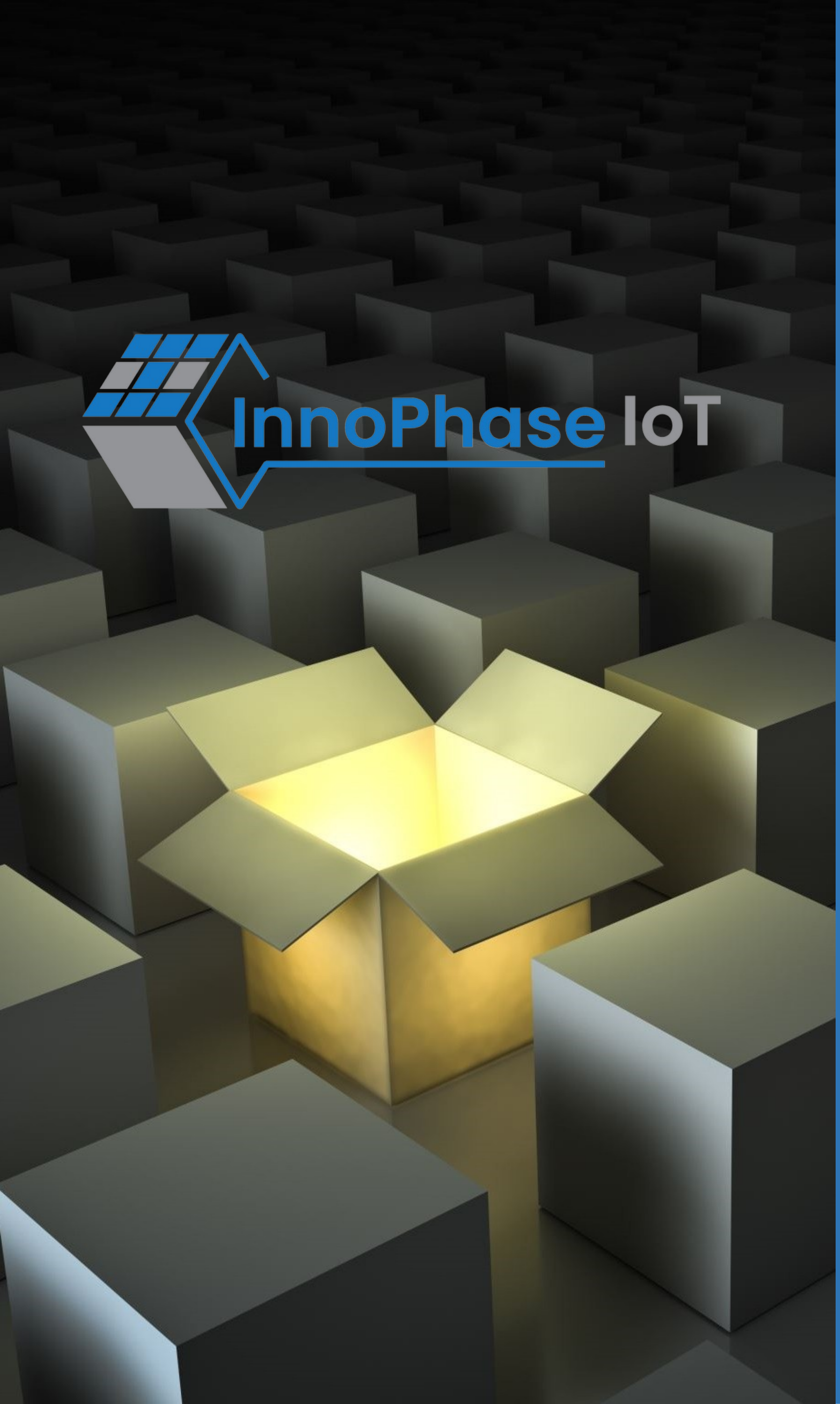
No	Peripheral	Status
1	GPIO	SUPPORTED (Limited test + Pin MUX)
2	UART	SUPPORTED (Except IrDA and RS-485)
3	SDIO	SUPPORTED
4	I2C	SUPPORTED
5	RTC	SUPPORTED
6	uDMA	SUPPORTED
7	LPTIMER	SUPPORTED
8	SPI	SUPPORTED
9	xSPI	SUPPORTED
10	Dual Timer	SUPPORTED
11	WDT	SUPPORTED
12	I2S	SUPPORTED
13	COMPARATOR	SUPPORTED

No	Peripheral	Status
14	TCPWM	NOT SUPPORTED
15	HKADC	NOT SUPPORTED
16	CLK	NOT SUPPORTED
17	PDM	NOT SUPPORTED
18	OTP	NOT SUPPORTED
19	SYS_PM	NOT SUPPORTED

ESP-IDF Compatible APIs

No	ESP API	INPH API
1	esp_wifi_init()	inph_wifi_init()
2	esp_wifi_deinit()	inph_wifi_deinit()
3	esp_wifi_start()	inph_wifi_start()
4	esp_wifi_stop()	inph_wifi_stop()
5	esp_wifi_set_mode()	inph_wifi_set_mode()
6	esp_wifi_set_config()	inph_wifi_set_config()
7	esp_wifi_connect()	inph_wifi_connect()
8	esp_wifi_disconnect()	inph_wifi_disconnect()
9	esp_wifi_scan_start()	inph_wifi_scan_start()
10	esp_wifi_scan_stop()	inph_wifi_scan_stop()
11	esp_wifi_scan_get_ap_num()	inph_wifi_scan_get_ap_num()
12	esp_wifi_scan_get_ap_records()	inph_wifi_scan_get_ap_records()
13	esp_wifi_clear_ap_list()	inph_wifi_clear_ap_list()
14	esp_wifi_set_ps()	inph_wifi_set_ps()
15	esp_wifi_get_ps()	inph_wifi_get_ps()
16	esp_wifi_set_default_wifi_sta_handlers()	inph_wifi_set_default_wifi_sta_handlers()

No	ESP API	INPH API
17	esp_wifi_set_default_wifi_ap_handlers()	inph_wifi_set_default_wifi_ap_handlers()
18	esp_wifi_clear_default_wifi_driver_and_handlers()	inph_wifi_clear_default_wifi_driver_and_handlers()
19	esp_wifi_destroy_if_driver()	inph_wifi_destroy_if_driver()
20	esp_wifi_get_if_mac()	inph_wifi_get_if_mac()
21	esp_wifi_register_if_rxcb()	inph_wifi_register_if_rxcb()



SQA Reports

Neon SQA Test Status – USDK-1.1.1-2026015-rc7.3

April 10, 2026

InnoPhase IoT

Wi-Fi STA Peak Tput – 11be

- Overall 11be and 11ax tput with Asus RT-BE86U is low compared to 11ax tput observed with Asus RT-AX86U PRO.
- In 11be mode, Tx frames from NeonB0 are sent at 11ax MCS-9 and Rx frames are received at 11be MCS-9

Mode – Security	Test AP	UDP DL	UDP UL	TCP DL	TCP UL
11be – WPA3-SAE	Asus RT-BE86U	36.8	40.3	32.8	28.4
11ax – WPA3-SAE	Asus RT-BE86U	38.2	38.7	29.5	26.9
11ax – WPA3-SAE	Asus RT-AX86U PRO	48.7	44.5	35.6	36.0

- DUT: **INP3117 Neon B0PP14**. Test duration: 60 secs. Tput in Mbps.
- Build: USDK 1.1.1-202615_rc7.3 (NPU: git-815500a; APU: b70a0313)
- Test AP: Asus RT-AX86U PRO; RSSI: -30 dBm; Channel 6
- Test Setup: Cabled inside RF Shield box; Clean channel
- SQA Test Report: April7_neon_b0_USDK1.1.1-202615_rc7.1_Neon_Consolidated_test_results_iperf3_server.xlsx

Wi-Fi STA Peak Tput – 11ax

- 11g/n/ax UDP/TCP UL tput increased by ~30% and 11n/ax TCP DL increased by ~20% compared to Rel6.

Mode – Security	UDP DL	UDP UL	TCP DL	TCP UL
11ax – Open	49.0	49.7	35.3	39.6
11ax – WPA2-PSK	48.5	43.7	34.5	37.1
11ax – WPA3-SAE	48.0	43.4	34.3	37.2

- DUT: **INP3117 Neon B0PP14**. Test duration: 60 secs. Tput in Mbps.
- Build: USDK 1.1.1-202615_rc7.3 (NPU: git-815500a; APU: b70a0313)
- Test AP: Asus RT-AX86U PRO; RSSI: -30 dBm; Channel 6
- Test Setup: Cabled inside RF Shield box; Clean channel
- SQA Test Report: April7_neon_b0_USDK1.1.1-202615_rc7.1_Neon_Consolidated_test_results_iperf3_server.xlsx

Wi-Fi STA Peak Tput – All modes

Open Security	UDP DL	UDP UL	TCP DL	TCP UL
11ax	49.8	46.1	34.2	32.0
11n	49.8	43.6	35.1	34.9
11g	28.0	20.9	16.3	17.7
11b	7.5	6.7	6.1	6.1
WPA2-PSK	**UDP DL**	**UDP UL**	**TCP DL**	**TCP UL**
11ax	49.8	42.5	32.7	36.0
11n	49.8	42.3	35.5	32.3
11g	27.8	20.9	14.7	12.3
11b	7.9	7.0	6.6	6.2
WPA3-SAE	**UDP DL**	**UDP UL**	**TCP DL**	**TCP UL**
11ax	48.7	44.5	35.6	36.0
11n	48.9	43.1	35.2	35.5
11g	27.8	23.8	15.2	15.4
11b	7.6	6.5	5.5	5.6

Wi-Fi STA Peak Tput – Rel7 vs Rel6

EVK - NEB0PP14. Asus RT86u Pro - Cabled Setup; RSSI: -30 dBm

Build details	802.11 mode WPA2 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11ax	**49.8**	**42.5**	**32.7**	**36.0**
RC6.3		48.1	32.5	32.2	27.0
Throughput % delta on RC7.1 compared to RC6.3		104%	131%	102%	133%

Build details	802.11 mode WPA3 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11ax	**48.7**	**44.5**	**35.6**	**36.0**
RC6.3		48.2	32.2	32.1	26.1
Throughput % delta on RC7.1 compared to RC6.3		101%	138%	111%	138%

Build details	802.11 mode WPA2 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11n	**49.8**	**42.3**	**35.5**	**32.3**
RC6.3		49.7	30.1	30.1	27.1
Throughput % delta on RC7.1 compared to RC6.3		100%	141%	118%	119%

Build details	802.11 mode WPA3 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11n	**48.9**	**43.1**	**35.2**	**35.5**
RC6.3		48.8	30.7	30.3	25.6
Throughput % delta on RC7.1 compared to RC6.3		100%	140%	116%	139%

Build details	802.11 mode WPA2 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11g	**27.8**	**20.9**	**14.7**	**12.3**
RC6.3		27.5	19.5	16.3	10.9
Throughput % delta on RC7.1 compared to RC6.3		101%	107%	90%	113%

Build details	802.11 mode WPA3 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11g	**27.8**	**23.8**	**15.2**	**15.4**
RC6.3		27.5	19.3	15.1	10.2
Throughput % delta on RC7.1 compared to RC6.3		101%	123%	101%	151%

Build details	802.11 mode WPA2 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11b	**7.9**	**7.0**	**6.6**	**6.2**
RC6.3		7.8	7.0	6.4	6.2
Throughput % delta on RC7.1 compared to RC6.3		101%	100%	103%	100%

Build details	802.11 mode WPA3 Security	UDP DL	UDP UL	TCP DL	TCP UL
RC7.1	11b	**7.6**	**6.5**	**5.5**	**5.6**
RC6.3		7.6	7.0	6.6	6.2
Throughput % delta on RC7.1 compared to RC6.3		100%	93%	83%	90%

Wi-Fi STA Rx Fixed Rate – 11be

Gaurd Interval	Rx MCS on NeonB0	Tx MCS Fixed rate commands on Asus AP	INP3119 B0PP17 UDP DL Peak Tput (Mbps) WPA3-SAE 11be Fixed rate Asus RT-BE88U	INP3119 B0PP17 UDP DL Peak Tput (Mbps) WPA3-SAE 11ax Fixed rate Asus RT-BE88U	INP3119 B0PP17 UDP DL Peak Tput (Mbps) WPA3-SAE 11ax Fixed rate Asus RT-AX86U PRO	PASS Criteria ESP32 C6 11ax RX Fixed Asus RT-AX86U PRO
0.8 us	MCS0	wl 2g_rate -t 0 -i 0 : 8.6 Mbps	3.7	3.9	7.2	7.1
1.6 us	MCS0	wl 2g_rate -t 0 -i 1 : 8.1Mbps	3.5	Not Run	6.8	6.7
3.2 us	MCS0	wl 2g_rate -t 0 -i 2 : 7.3 Mbps	3.2	Not Run	6.1	6.0
0.8 us	MCS1	wl 2g_rate -t 1 -i 0 : 17.2 Mbps	7.4	7.6	14.7	14.2
1.6 us	MCS1	wl 2g_rate -t 1 -i 1 : 16.3 Mbps	6.9	Not Run	13.6	13.5
3.2 us	MCS1	wl 2g_rate -t 1 -i 2 : 14.6 Mbps	6.3	Not Run	12.6	12.2
0.8 us	MCS2	wl 2g_rate -t 2 -i 0 : 25.7 Mbps	10.8	11.6	21.9	21.5
1.6 us	MCS2	wl 2g_rate -t 2 -i 1 : 24.4 Mbps	10.9	Not Run	21.0	20.2
3.2 us	MCS2	wl 2g_rate -t 2 -i 2 : 21.9 Mbps	9.4	Not Run	18.5	18.1
0.8 us	MCS3	wl 2g_rate -t 3 -i 0 : 34.4 Mbps	15.5	15.0	29.6	28.6
1.6 us	MCS3	wl 2g_rate -t 3 -i 1 : 32.5 Mbps	13.9	Not Run	26.7	27.2
3.2 us	MCS3	wl 2g_rate -t 3 -i 2 : 29.3 Mbps	13.4	Not Run	23.8	24.4
0.8 us	MCS4	wl 2g_rate -t 4 -i 0 : 51.6 Mbps	23.3	22.9	44.3	42.6
1.6 us	MCS4	wl 2g_rate -t 4 -i 1 : 48.8 Mbps	22.0	Not Run	41.1	40.2
3.2 us	MCS4	wl 2g_rate -t 4 -i 2 : 43.9 Mbps	19.5	Not Run	37.5	36.3
0.8 us	MCS5	wl 2g_rate -t 5 -i 0 : 68.8 Mbps	13.2	10.0	43.0	56.4
1.6 us	MCS5	wl 2g_rate -t 5 -i 1 : 65 Mbps	7.8	Not Run	43.7	53.2
3.2 us	MCS5	wl 2g_rate -t 5 -i 2 : 58.5 Mbps	16.5	Not Run	36.5	48.0
0.8 us	MCS6	wl 2g_rate -t 6 -i 0 : 77.4 Mbps	25.5	28.8	44.5	63.3
1.6 us	MCS6	wl 2g_rate -t 6 -i 1 : 73.1 Mbps	13.2	Not Run	44.2	60.0
3.2 us	MCS6	wl 2g_rate -t 6 -i 2 : 65.8 Mbps	8.9	Not Run	45.5	53.4
0.8 us	MCS7	wl 2g_rate -t 7 -i 0 : 86 Mbps	29.5	28.3	44.4	62.0
1.6 us	MCS7	wl 2g_rate -t 7 -i 1 : 81.3 Mbps	25.0	Not Run	44.9	63.1
3.2 us	MCS7	wl 2g_rate -t 7 -i 2 : 73.1 Mbps	10.9	Not Run	44.5	59.5
0.8 us	MCS8	wl 2g_rate -t 8 -i 0 : 103.2 Mbps	32.3	35.5	43.9	TBD
1.6 us	MCS8	wl 2g_rate -t 8 -i 1 : 97.5 Mbps	33.2	Not Run	45.2	TBD
3.2 us	MCS8	wl 2g_rate -t 8 -i 2 : 87.8 Mbps	27.5	Not Run	44.8	TBD
0.8 us	MCS9	wl 2g_rate -t 9 -i 0 : 114.7 Mbps	33.9	37.1	28.5	TBD
1.6 us	MCS9	wl 2g_rate -t 9 -i 1 : 108.3 Mbps	30.5	Not Run	43.9	TBD
3.2 us	MCS9	wl 2g_rate -t 9 -i 2 : 97.5 Mbps	32.1	Not Run	44.7	TBD

- With Asus RT-BE88U AP (Wi-Fi 7) - Both 11be and 11ax Rx fixed rates are yielding lower throughput for all data rates.
- With Asus RT-AX86U AP (Wi-Fi 6) - This issue is not observed, but lower throughput is observed using higher data rate due to a know issue (UDL DL drastically decreased with BW >50M)

Wi-Fi STA Connectivity

- Connectivity with all supported security types, reconnection and soft roaming are tested working. No regressions observed.

Test Case	Results	Comments
Connectivity – Open Sec	PASS	
Connectivity – WPA-PSK-AES	PASS	
Connectivity – WPA2-PSK-AES – PMF Disabled	PASS	
Connectivity – WPA2-PSK-AES – PMF Capable	PASS	STA connects with PMF
Connectivity – WPA2-PSK-AES – PMF Required	PASS	STA connects with PMF
Connectivity – WPA3-SAE	PASS	Higher connection latency (3-5 secs)
Connectivity – WPA/WPA2	PASS	STA connects with WPA2-PSK security
Connectivity – WPA2/WPA3	PASS	STA connects with WPA3-SAE security
Reconnection – AP config change	PASS	Reconnection successful on Channel, SSID, Passphrase and security change. DUT delays Tx of deauth frames during reconnection.
Reconnection – Link Loss	PASS	STA reconnects after link loss when AP is back online
WPA3-SAE with various length of passphrase	PASS	
WPA3-SAE Groups	PASS	
Soft roam between two APs with WPA3-SAE	PASS	
Soft roam between two APs with WPA2-PSK and WPA3-SAE	PASS	
Soft roam between two APs with WPA2/WPA3-SAE to WPA2	PASS	

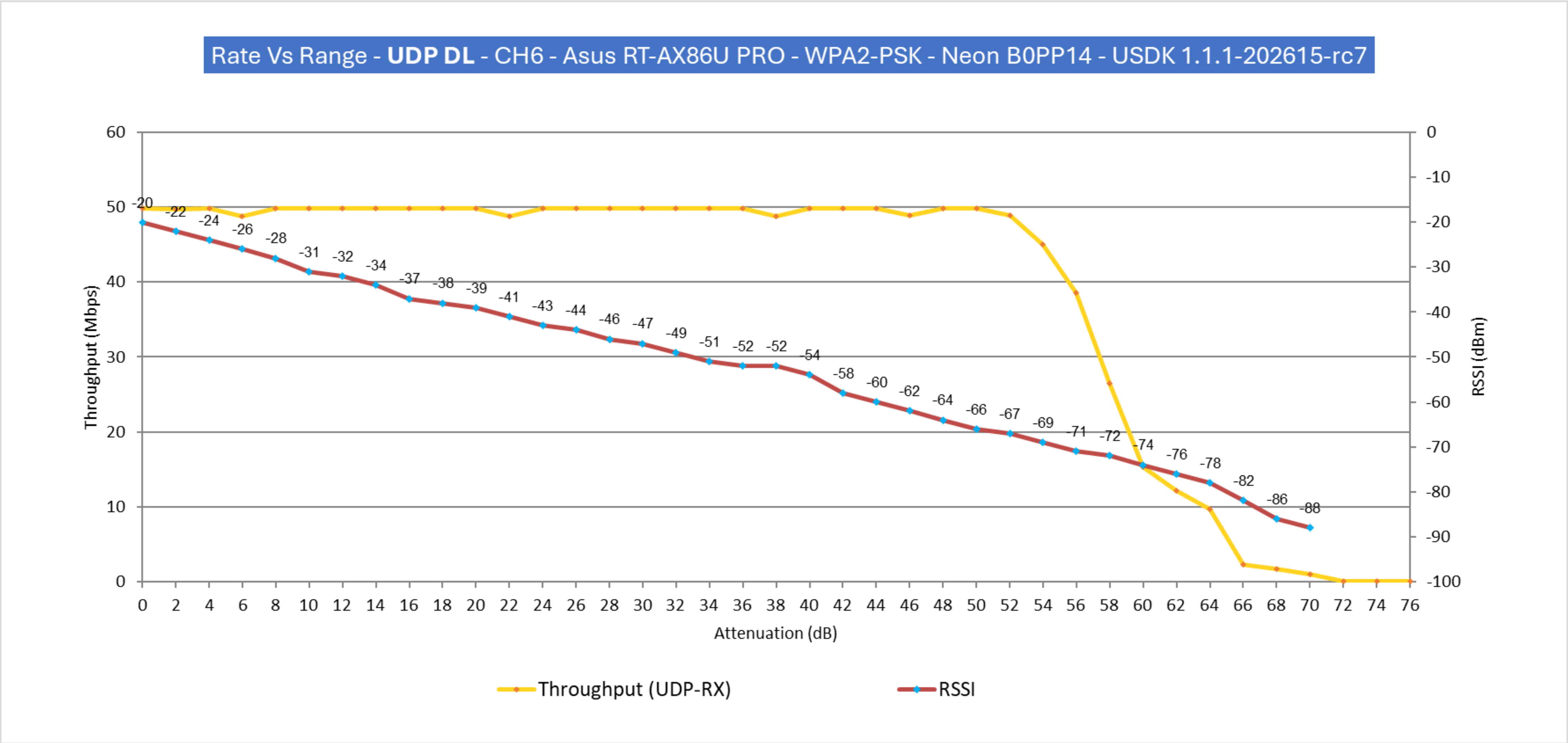
RvR Summary - WPA2-PSK

- Near range UL performance has reduced compared to Rel6
- Stalls are no longer observed during TCP DL traffic at lower RSSIs.
- Near range performance is lower for UDP/TCP DL and TCP UL traffic.

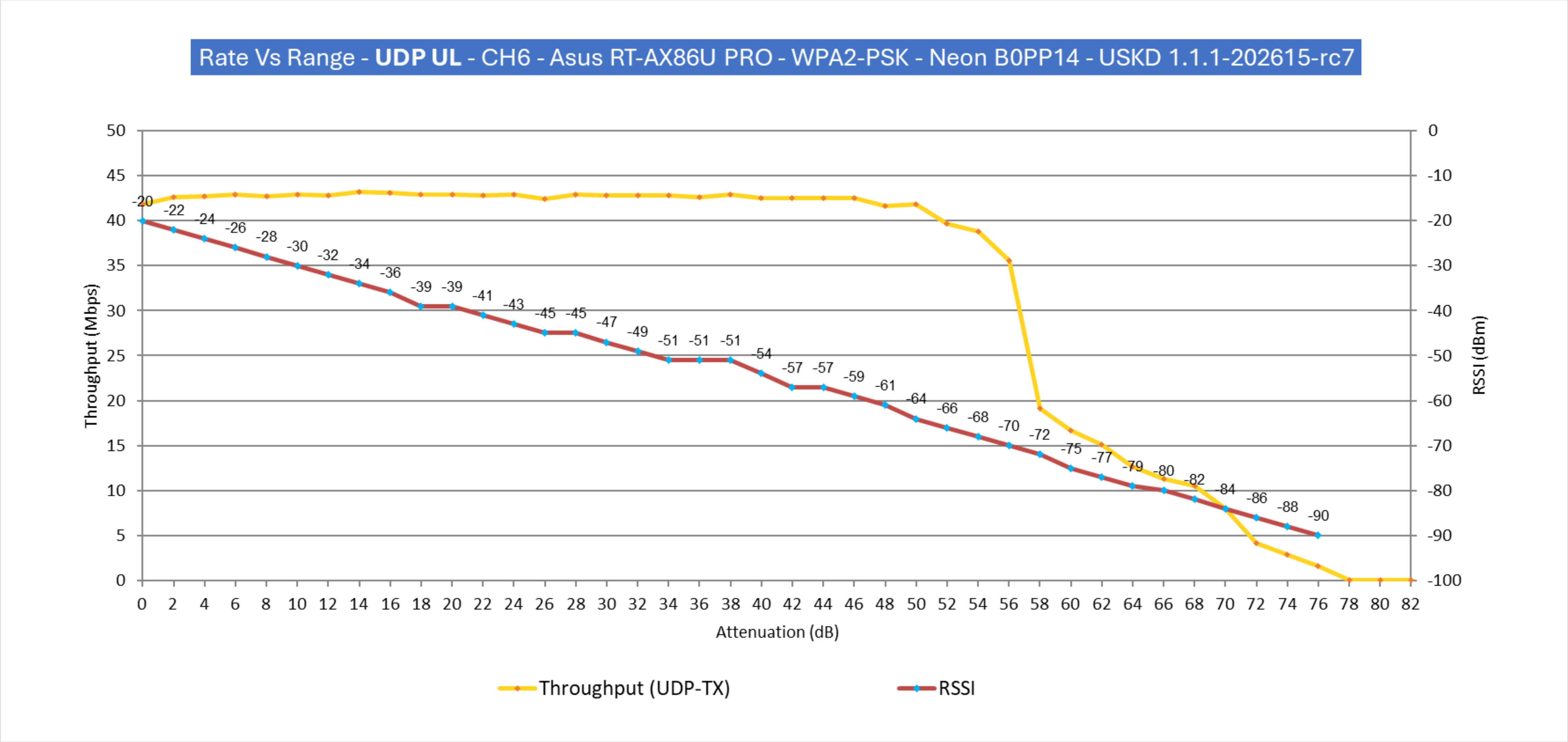
WPA2-PSK	USDK-1.1.1-rc7		USDK-1.1.1-rc6	
	Near Range	Sensitivity	Near Range	Sensitivity
UDPDL	-67	-88	-68	-86
UDPUL	-64	-90	-79	-92
TCPDL	-66	-92	-68	-90
TCPUL	-67	-92	-75	-90

- DUT: **INP3117 Neon B0PP14**. RSSI in dBm.
- Build: USDK 1.1.1-202615_rc7.1 (NPU: git-815500a; APU: 2915f8ba)
- Test AP: Asus RT-AX86U PRO; Channel 6
- Test Setup: Cabled inside RF Shield box; Clean channel
- SQA Test Report: April7_neon_b0_USDK1.1.1-202615_rc7.1_Neon_Consolidated_test_results_iperf3_server.xlsx

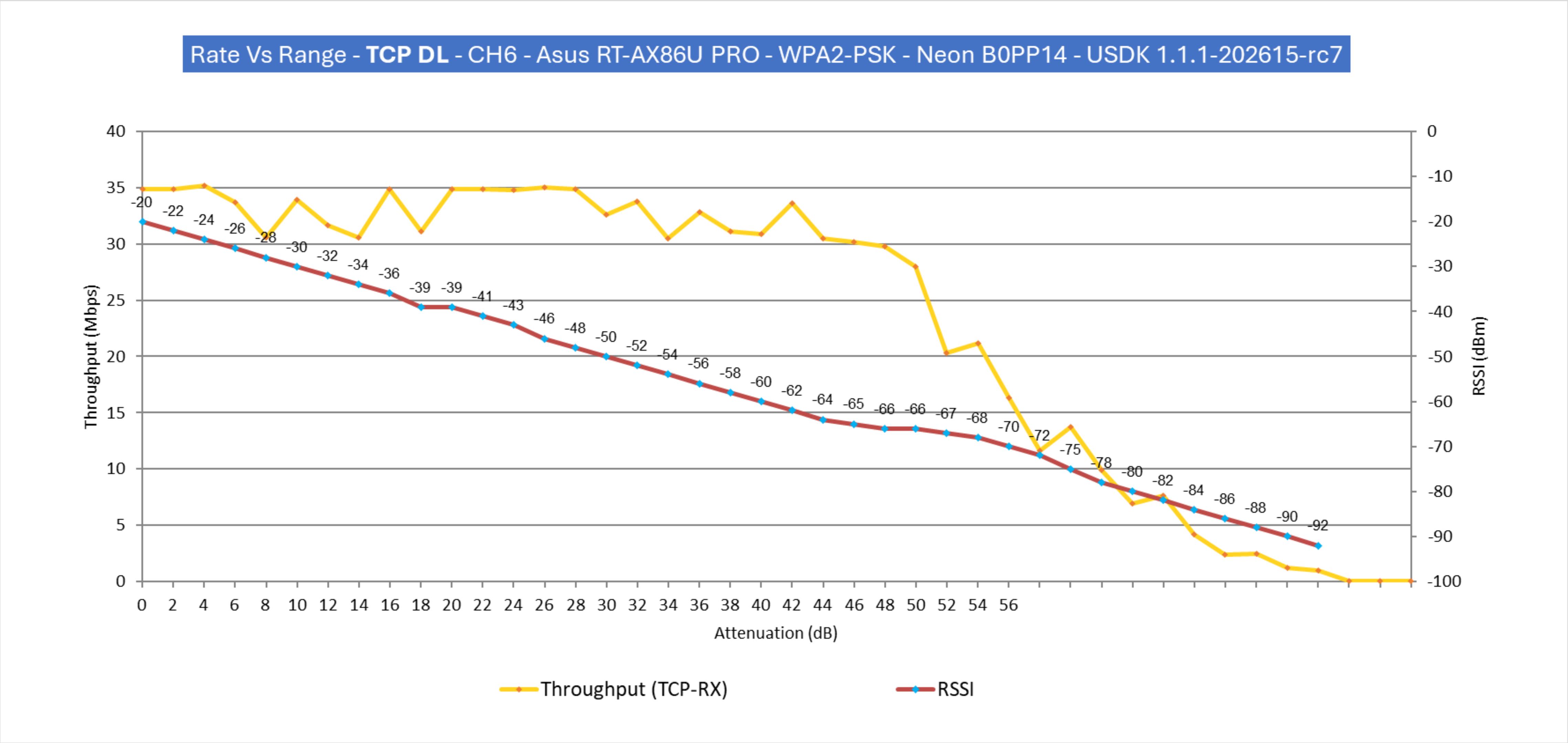
RvR – UDP DL – WPA2-PSK



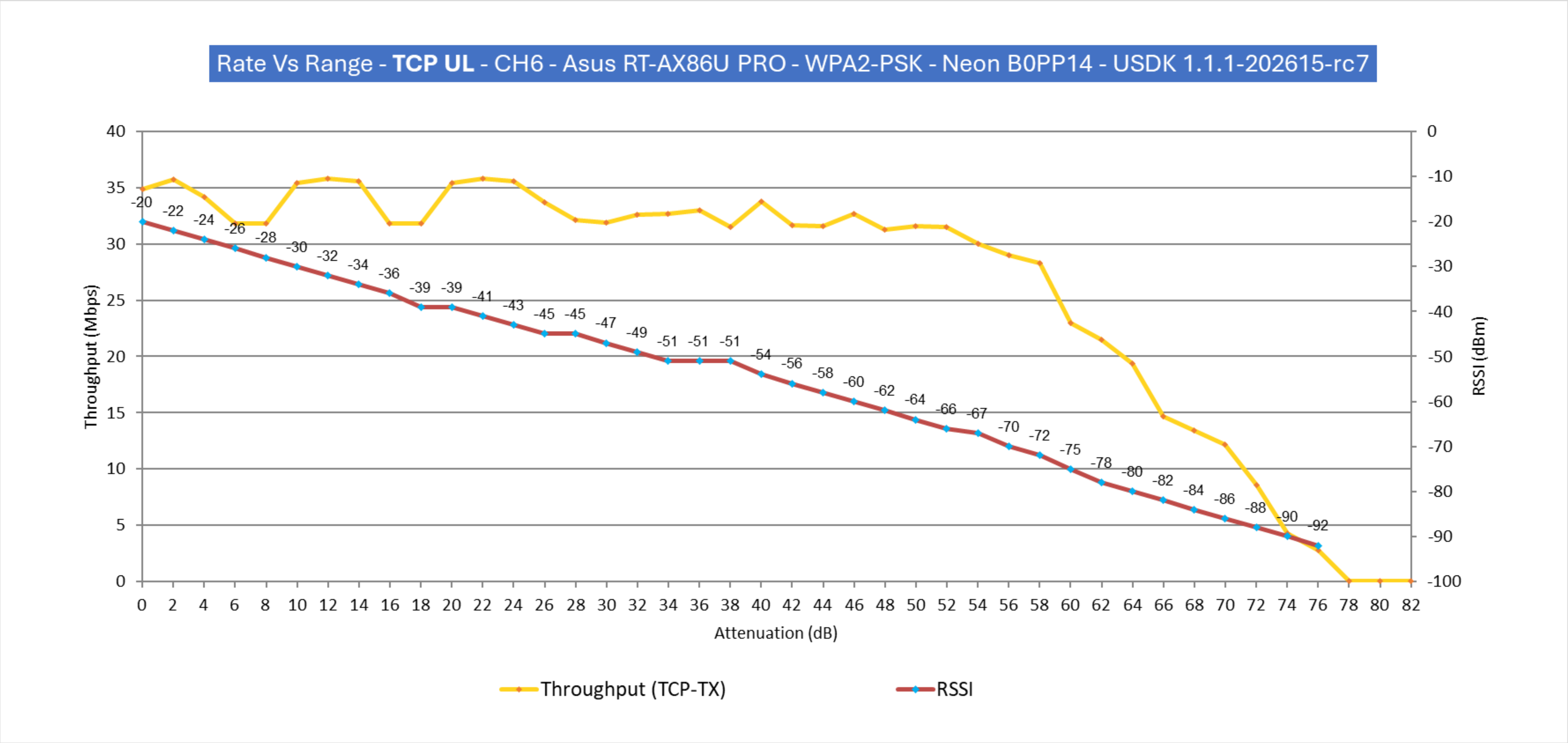
RvR – UDP UL – WPA2-PSK



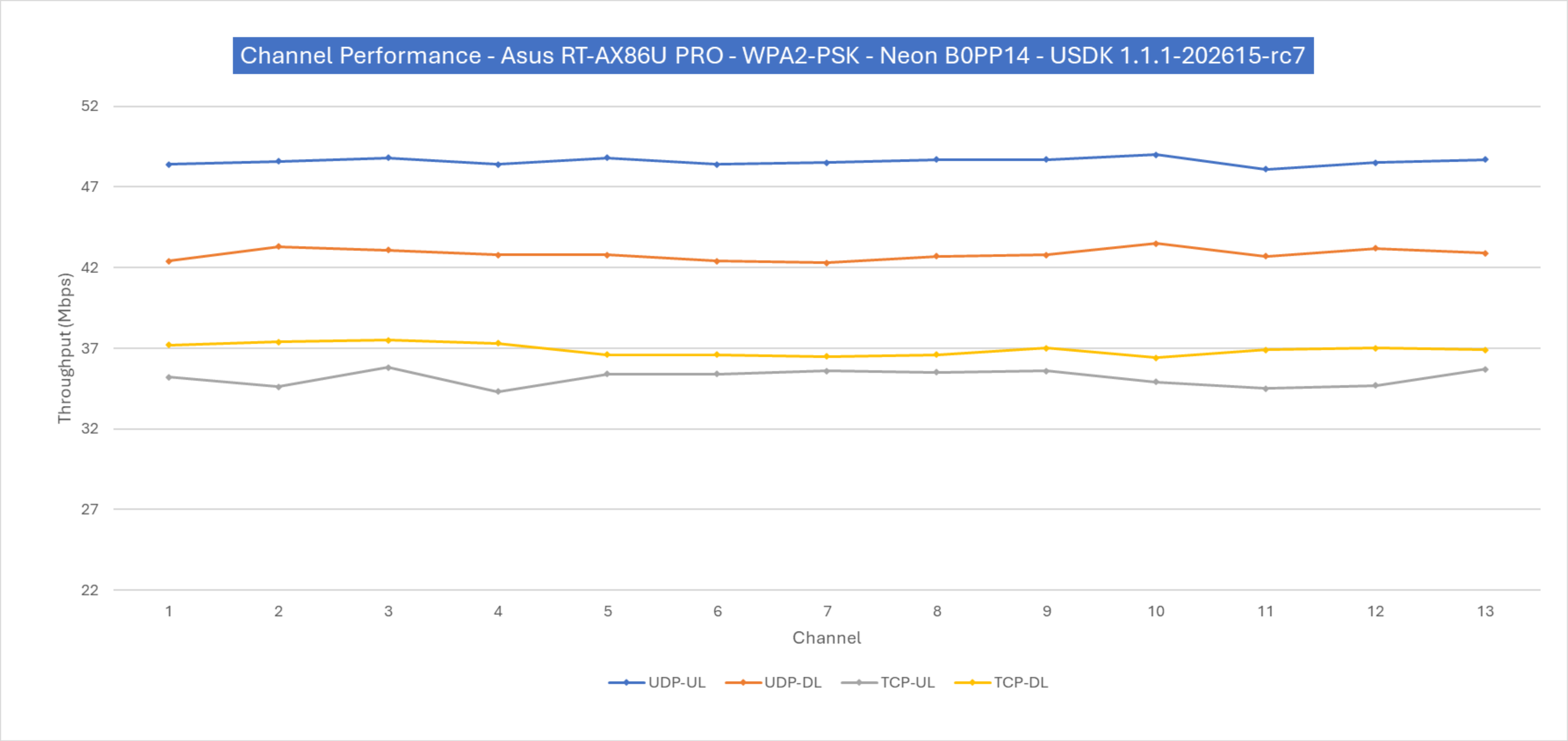
RvR – TCP DL – WPA2-PSK



RvR – TCP UL – WPA2-PSK

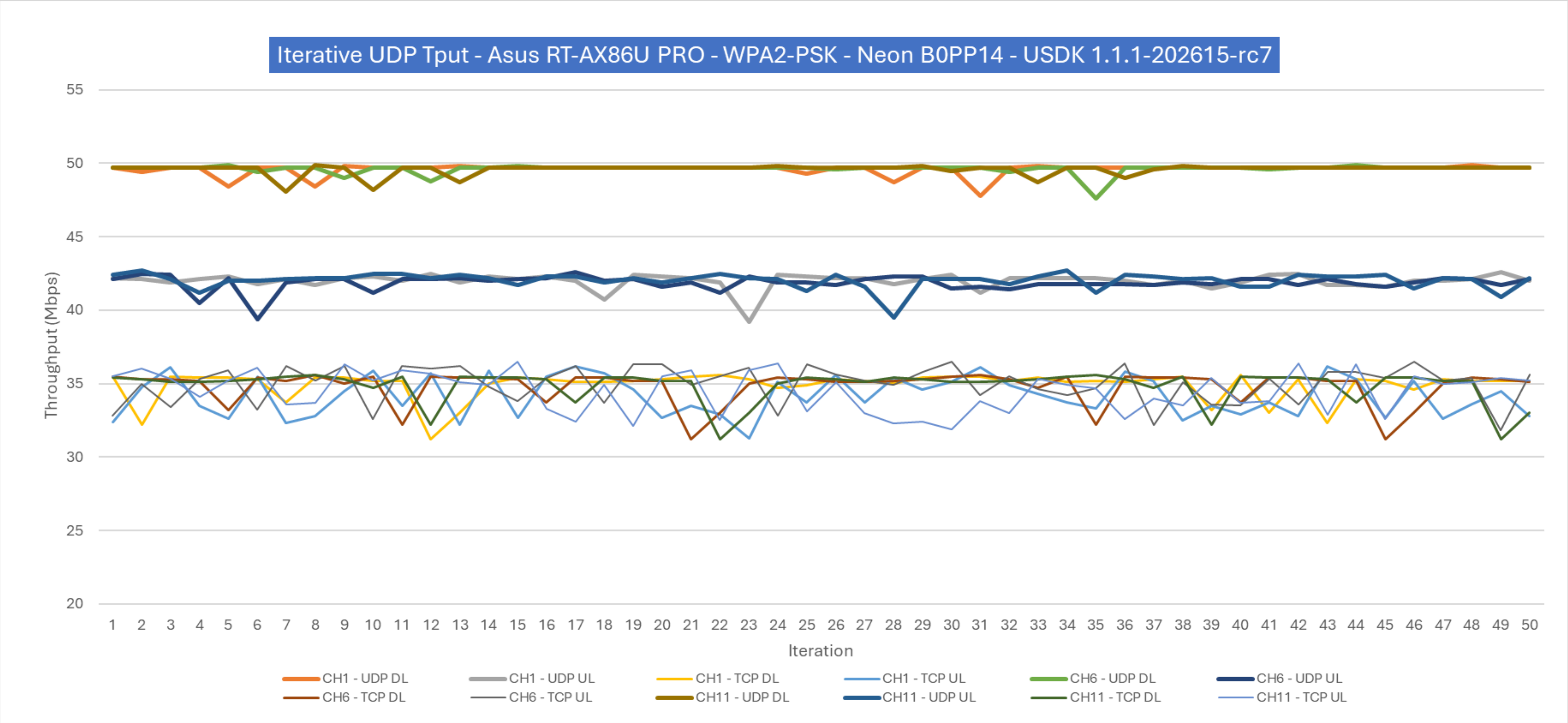


Channel Performance – WPA2-PSK



Repeated Tput – WPA2-PSK

- Intermittent data stalls are not observed in TCP DL tests with Rel7
- Throughput variation observed across iterations, higher in TCP. DUT reset performed for each iteration



Wi-Fi STA Low Power – Rel7

- Rel7 shows no low-power regression; results are at par with Rel6

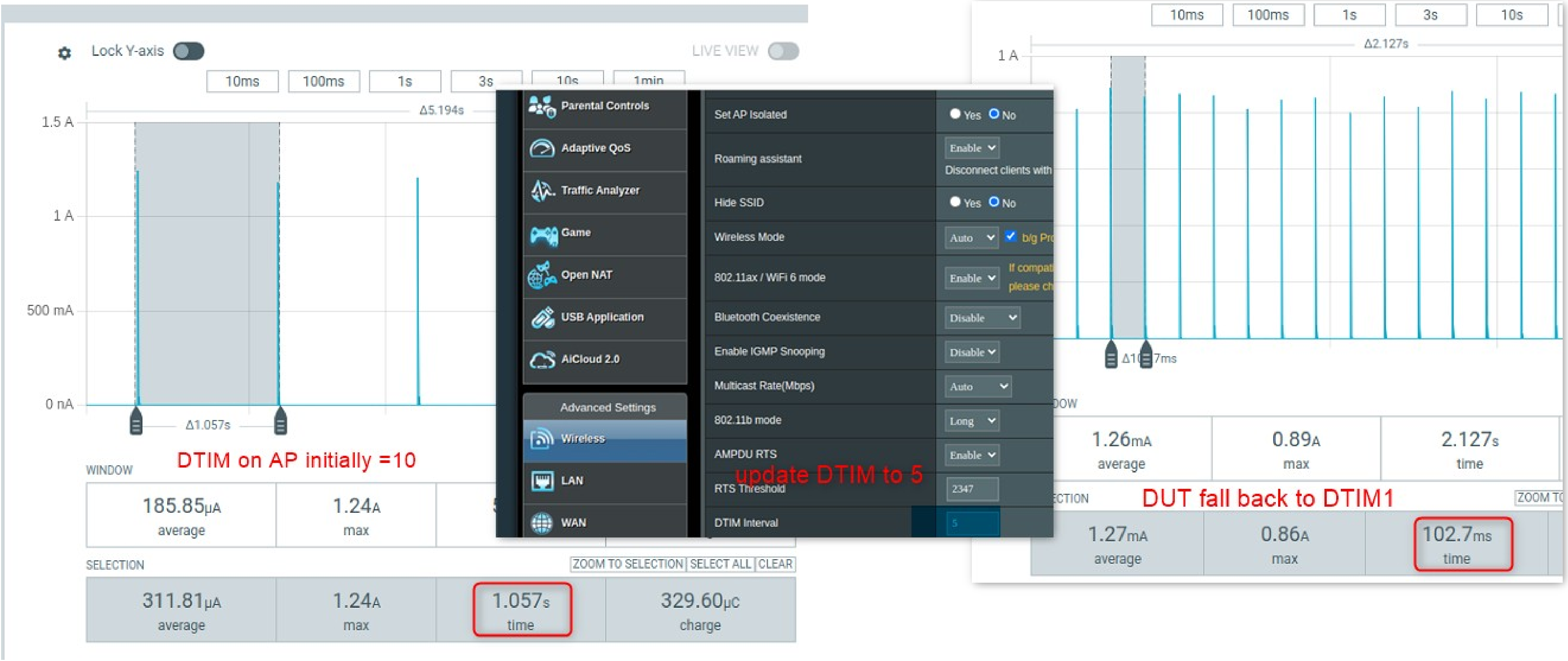
Setup details	DTIM configured on AP	RC6.3 Avg DTIM Current (µA)	**RC7.1 Avg DTIM Current (µA)**
Build: neon_b0_USDK1.1.1-202615_rc7.1 **DUT:** Neon B0PP19 **Test AP:** Asus RT-AX86U PRO; **RSSI:** -34 dBm. **Multicast traffic disabled** **Environment:** Cabled inside RF Shield box; Clean channel 1, No beacon miss.	DTIM1	1220	**1230**
	DTIM2	650	**646**
	DTIM3	451	**449**
	DTIM5	302	**303**
	DTIM10	193	**190**
	DTIM30	106	**108**
	DTIM100	76	**75**

Application	Wi-Fi	RC6.3 Suspend Current (µA)	**RC7.1 Suspend Current (µA)**
sysPM_suspend.elf	No	30	**30**
sysPM_UDP.elf - DTIM10	yes	60	**60**

Wi-Fi STA Low Power - Observations

New Observation:

- DUT fails to honor updated DTIM on AP and reverts to DTIM1 post-reconnect – JIRA USDK-284
  - The DUT adopts the updated DTIM value from the AP only after a reset



Wi-Fi STA Interoperability – 11be

- No critical issues observed with the five 11be routers tested.
- Overall traffic is low with BE5100 and RT-BE88U routers. UDP DL is low with 90M BW.

Test AP	Wi-Fi Chipset	RSSI	UDP DL	UDP UL	TCP DL	TCP UL
TP-Link BE5100	MediaTek MT7991AV	-34 dBm	34 (25)	22	24	19
Asus RT-BE82U	Broadcom BCM6765	-35 dBm	40 (7)	40	32	32
TP-Link BE6500	Broadcom BCM6766	-37 dBm	40 (13)	43	33	31
Asus RT-BE86U	Broadcom BCM4916	-30 dBm	27 (10)	36	20	25
Asus RT-BE88U	Broadcom BCM43684	-30 dBm	36 (8)	42	31	30

- DUT: INP3119 Neon B0PP12. Tput in Mbps. WPA3-SAE security
- Test Setup: Radiated inside RF Shield box; Clean channel
- UDP BW: 50M (90M)
- SQA Test Report: Neon_IOP_Test_Results.xlsx

Wi-Fi STA Interoperability – 11ax

- NPU crash (EVENT_WIFI_TX_UNDERRUN) seen during UDP DL traffic with TP-Link AP.
- Data stalls are seen during TCP traffic with several APs. Low tput observed with Asus and Huawei APs.

Test AP	Wi-Fi Chipset	RSSI	UDP DL	UDP UL	TCP DL	TCP UL
ASUS RT-AX86U Pro	Broadcom BCM43684	-36 dBm	49 (9)	45	35	33
TP-Link AX6000	Broadcom BCM43684	-34 dBm	47 (0)	15	0	0
ASUS RT-AX88U	Broadcom BCM43684	-40 dBm	44 (17)	34	0	14
Mercusys MR80X	MediaTek MT7981BA	-36 dBm	46 (7)	39	32	0
Huawei WS7200 AX3	Hisilicon Hi1152	-35 dBm	49 (1)	22	18	18
Xiaomi MI RA72 AX6000	Qualcomm IPQ5018	-30 dBm	39 (6)	40	0	0
NetGear AXE11000	Broadcom BCM43684	-27 dBm	49 (2)	45	35	33

- DUT: INP3119 Neon B0PP12. Tput in Mbps. WPA3-SAE security
- Test Setup: Radiated inside RF Shield box; Clean channel
- UDP BW: 50M (90M)
- SQA Test Report: Neon_IOP_Test_Results.xlsx

Wi-Fi STA Stability – 11ax, 24 hours

- Overall stability has improved in Rel7.
- TCP UL and UDP DL traffic is unstable in noisy channels.

24 Hours iPerf Traffic Stability

Test Scenario	Test duration	Channel	Neon B0 Board	Test AP	Result	Comments
TCP DL	24 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 35.8 Mbps
TCP UL	24 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 34.8 Mbps
UDP DL (50M BW)	24 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	FAIL	Out of memory failure in 80 mins
UDP UL (50M BW)	24 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 45.4 Mbps
TCP DL	24 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	PASS	Tput at end of test – 1.3 Mbps
TCP UL	24 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	FAIL	Data stalls in 47 mins. STA in hung state.
UDP DL (50M BW)	24 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	FAIL	Out of memory failure after 10.5 hours
UDP UL (50M BW)	24 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	PASS	Tput at end of test – 5.5 Mbps

- Test AP Config: WAP3-SAE, Channel 6, DTIM-1
- Noisy channel condition is case-3 (high interference) environment; BLR office/Lab
- SQA Test Report: SQA_WiFi_BLE_Peripheral_Stability_Summary_April12_2026.xlsx

Wi-Fi STA Stability – 11ax, 2 hours

2 Hours iPerf Traffic Stability

Test Scenario	Test duration	Channel	Neon B0 Board	Test AP	Result	Comments
TCP DL	2 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 35.6 Mbps
TCP UL	2 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 34.8 Mbps
UDP DL (50M BW)	2 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 47.7 Mbps
UDP UL (50M BW)	2 Hours	**Cabled, Clean**	INP6120 RK06	Asus RT-AX86U PRO	PASS	Tput at end of test – 45.4 Mbps
TCP DL	2 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	PASS	Tput at end of test – 2.3 Mbps
TCP UL	2 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	FAIL	Data stalls in 47 mins. STA in hung state.
UDP DL (50M BW)	2 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	PASS	Tput at end of test – 1.4 Mbps
UDP UL (50M BW)	2 Hours	**OTA, Noisy**	INP3117 B0PP12	Asus RT-AX86U PRO	PASS	Tput at end of test – 4.3 Mbps

- Test AP Config: WAP3-SAE, Channel 6, DTIM-1
- Noisy channel condition is case-3 (high interference) environment; BLR office Lab
- SQA Test Report: SQA_WiFi_BLE_Peripheral_Stability_Summary_April12_2023.xlsx

Wi-Fi STA Stability – 11be, 2 hours

- NPU crash (EVENT_WIFI_TX_UNDERRUN) seen during UDP DL traffic with TP-Link AP.

2 Hours iPerf Traffic Stability

Test Scenario	Test duration	Channel	Neon B0 Board	Test AP	Result	Comments
TCP DL	2 Hours	**Cabled, Clean**	INP3119 B0PP12	Asus RT-BE86U	PASS	Tput at end of test – 29.3 Mbps
TCP UL	2 Hours	**Cabled, Clean**	INP3119 B0PP12	Asus RT-BE86U	PASS	Tput at end of test – 30.3 Mbps
UDP DL (50M BW)	2 Hours	**Cabled, Clean**	INP3119 B0PP12	Asus RT-BE86U	FAIL	NPU crash (EVENT_WIFI_UNDERRUN) in 40 mins
UDP UL (50M BW)	2 Hours	**Cabled, Clean**	INP3119 B0PP12	Asus RT-BE86U	PASS	Tput at end of test – 40.3 Mbps
TCP DL	2 Hours	**OTA, Noisy**	INP3119 B0PP12	Asus RT-BE86U	In-Progress	
TCP UL	2 Hours	**OTA, Noisy**	INP3119 B0PP12	Asus RT-BE86U	In-Progress	
UDP DL (50M BW)	2 Hours	**OTA, Noisy**	INP3119 B0PP12	Asus RT-BE86U	FAIL	NPU crash (EVENT_WIFI_UNDERRUN) in 17 mins
UDP UL (50M BW)	2 Hours	**OTA, Noisy**	INP3119 B0PP12	Asus RT-BE86U	In-Progress	

- Test AP Config: WAP3-SAE, Channel 6, DTIM-1
- Noisy channel condition is case-3 (high interference) environment; BLR office lab
- SQA Test Report: [SQA_WiFi_BLE_Peripheral_Stability_Summary_April12_2026.xlsx]

BLE Exampe Apps with REF devices

Test App	Test Details	Android Pixel-9	iOS iPhone-15	IFX Devit CYW920829M2	Comments
nimble_beacon	non connectable advertisement	PASS	PASS	PASS	Advertisements seen
nimble_connection	Connectable advertisement and connection	PASS	PASS	FAIL	Connection failure with IFX
nimble_gatt_server	Connectable advertisement, connection and GATT operation - read/write/Notify/indicate	PASS	PASS	FAIL	Connection failure with IFX
nimble_security	Connectable advertisement, connection and GATT operation - read/write/Notify/indicate; HRS to read/write with security 6-digit pin enabled	PASS	PASS	FAIL	Connection failure; disconnect observed after passkey with IFX
nimble_scanner	Scans for the BLE devices and extract address, RSSI advertisement payload fields — device name, service UUIDs, manufacturer data, etc.	PASS	PASS	PASS	
nimble_external_adv_ADV_EXT_MAX_EVENTS	100 events for extended advertisement	PASS	FAIL	FAIL	Ext Adv not observed on iPhone and IFX devkit.
nimble_external_adv_ADV_SCANNABLE_LEGACY_EXT	Simple non-connectable scannable instance using legacy PUDs.	PASS	FAIL	PASS	Ext Adv not observed on iPhone.
nimble_external_adv_ADV_LEGACY_DURATION	Starts advertising instance with 5sec timeout and changing adv data pattern. Advertisements are observed.	PASS	PASS	PASS	New adv packets with MAC address change
nimble_external_adv_ADV_SCANNABLE_EXT	This is simple scannable instance that runs forever.	PASS	FAIL	FAIL	Ext Adv not observed on iPhone and IFX devkit.
nimble_external_adv_ADV_PERIODIC	Start periodic advertise packets	FAIL	FAIL	FAIL	Wrong MAC address observed in on Pixel. Scan failure on iPhone and IFX

- Test Setup: Office environment; noisy channel
- SQA Test Report: [neon_b0_USDK1.1.1-202615_rc7.1.xlsx]

BLE Peak Tput

- Tput with 1M PHY has significantly improved in Rel7.
- Disconnects are observed in LRS2 and 1M PHY. Data stalls and hangs are observed in LRS8 PHY.
- Tput in 1M PHY, small packets, decrease with increase in connection interval.

PHY	Packet Length	7.5 ms		50 ms		100 ms	
		TX	RX	TX	RX	TX	RX
1M	27 Bytes Min Length	312	312	232	232	144	144
2M		464	464	512	512	512	512
LRS2		-1	-1	-1	-1	-1	-1
LRS8		-1	-1	-1	-1	-1	-1

PHY	Packet Length	7.5 ms		50 ms		100 ms	
		TX	RX	TX	RX	TX	RX
1M	251 Bytes Max Length	536	536	688	688	592	592
2M		1344	1344	1416	1416	1448	1448
LRS2		-1	-1	-1	-1	-1	-1
LRS8		-1	-1	-1	-1	-1	-1

- DUT (Peripheral): Neon B0 NEB0PP20
- DUT (Central): Neon B0 NEA0PP52
- Test Setup: Cabled inside RF Shield box; Clean channel
- -1 indicate test failures (disconnect, crash, hangs etc.)

BLE Peak Tput – Rel7 vs Rel6

PHY	Build	Packet Length (Bytes)	Connection Interval 7.5ms		Connection Interval 50ms		Connection Interval 100ms	
			TX	RX	TX	RX	TX	RX
1M	USDK1.1.1-202613_rc6.3	27	256	256	80	80	40	40
	USDK1.1.1-202615_rc7.2		312	312	232	232	144	144
	Tput on rc7.2 as % of rc6.3		**121.9%**	**121.9%**	**290.0%**	**290.0%**	**360.0%**	**360.0%**
	USDK1.1.1-202613_rc6.3	251	512	512	480	480	360	360
	USDK1.1.1-202615_rc7.2		536	536	688	688	592	592
	Tput on rc7.2 as % of rc6.3		**104.7%**	**104.7%**	**143.3%**	**143.3%**	**164.4%**	**164.4%**

PHY	Build	Packet Length (Bytes)	Connection Interval 7.5ms		Connection Interval 50ms		Connection Interval 100ms	
			TX	RX	TX	RX	TX	RX
2M	USDK1.1.1-202613_rc6.3	27	472	472	496	496	488	496
	USDK1.1.1-202615_rc7.2		464	464	512	512	512	512
	Tput on rc7.2 as % of rc6.3		**98.3%**	**98.3%**	**103.2%**	**103.2%**	**104.9%**	**103.2%**
	USDK1.1.1-202613_rc6.3	251	1360	1360	1424	1424	1408	1416
	USDK1.1.1-202615_rc7.2		1344	1344	1416	1416	1448	1448
	Tput on rc7.2 as % of rc6.3		**98.8%**	**98.8%**	**99.4%**	**99.4%**	**102.8%**	**102.3%**

Networking Protocol Example apps

- All networking protocol example apps are tested working. No critical issues observed.
- README update needed for some of the example apps.

Example App	Test Details	Results	Comments
esp_http_client_example	Simple HTTP client request and response.	PASS	httpbin.org server used for verification
http_server simple	HTTPD server with URI handling. GET and POST command verification.	PASS	Curl commands used to GET, PUT, POST operations with client.py script
https_mbedtls	simple HTTPS client request that uses mbedTLS to establish a secure socket connection using the certificate bundle with two custom certificates added for verification.	PASS	SSL/TLS secure connection howsmyssl.com:443 used for verification
mqtt_ssl	Connects to the broker using ssl transport and as a demonstrate subscribes/unsubscribes and send a message on certain topic	PASS	mqtt broker mqtts://broker.emqx.io:8883 used for verification
mqtt_tcp	Connects to the broker using mqtt tcp transport and as a demonstrate subscribes/unsubscribes and send a message on certain topic.	PASS	mqtt broker mqtt://broker.emqx.io used for verification
mqtt_websocket	Connects to the broker over web sockets and demonstrate subscribes/unsubscribes and send a message on certain topic.	PASS	mqqt broker ws://broker.emqx.io:8083/mqtt used for verification
mqtt_websocket_secure	Connects to the mqtt broker over secure websockets and demonstrates subscribes/unsubscribes and send a message on certain topic.	PASS	mqtt broker wss://broker.emqx.io:8084/mqtt used for verification
sntp	Demonstrates the use of LwIP SNTP module to obtain time from Internet servers	PASS	pool.ntp.org internet server used to NTP get time for verification
tcp_transport_client	Demonstrates the TCP transport client send and receive messages.	PASS	Linux netcat server (nc -l -s <Ipaddress> -p <port>) used for verification

Software SWD reset and bootarg support

- Clear flash issue not observed in Rel7. No regressions observed
- Soft reset does not work on INP3119 EVK and INP6120 RK EVB
- APU crash (EXCEPTION 5) is observed sometimes after clear flash operation

Tests results with multiple programmer or FTDI boards (2) connected to PC at a time	INP3000 v5.0 or v6.0 + Neon INP3117 B0PP with J28 jumper neon_b0_USDK1.1.1-202615_rc7.1			Neon B0 INP6120 neon_b0_USDK1.1.1-202615_rc7.1			INP3000 v5.0 or v6.0 + Neon INP3119 B0PP with J28 jumper neon_b0_USDK1.1.1-202615_rc7.1		
	Clear Flash	Program flash	Soft Reset	Clear flash	Program flash	Soft reset	Clear Flash	Program flash	Soft Reset
Download tool Linux	Pass	Pass	Pass	Pass	Pass	Fail USDK-245	Pass	Pass	Fail USDK-271
Boot.py commands on Linux	Pass	Pass	Pass	Pass	Pass		Pass	Pass	
Download tool Windows	Pass	Pass	Pass	Pass	Pass		Pass	Pass	
Boot.py commands on Windows	Pass	Pass	Pass	Pass	Pass		Pass	Pass	

- Boot arguments support tested working using rc7Download tool and boot.py script on Windows and Linux OS.
- Programmed boot arguments can be seen in the NPU console logs. Also verified using sniffer logs (wifi.sounding, wifi.he, wifi.ht enable/disable works)
- Existing issue: Hwaddr, suspend, ssid and passphrase boot args do not take effect.

Peripherals Test Results – SPI

SPI-0 and SPI-1 Test Cases	Result	Remark
0 - SPI getVersion_Test	PASS	Executed using Automated test script and manual test.
1 - SPI Master write data to Flash	PASS	Manual test.
2 - SPI Master read data from Flash	PASS	Manual test.
3 - SPI Master Full Duplex test	PASS	Manual test.
4 - SPI Master 1 page write and read, 16 bit	PASS	Manual test.
5 - SPI Master Full Duplex 16 bit	PASS	Manual test.
6 - SPI Master DMA Tx	PASS	Manual test.
7 - SPI Master DMA Rx	PASS	Manual test.
8 - SPI Slave send data test	PASS	Executed using Automated test script
9 - SPI Slave receive data test	PASS	Executed using Automated test script
a - SPI Set pins HiZ	PASS	Manual test.

- DUT: Neon B0PP16
- SQA Test Report: NEON_B0_RC7.1_Result_7thApril2026.xlsx

Peripherals Test Results – I2S

I2S Test Cases	Sub-Module Test Selection	Result	Remark
0 - I2S initialization test	I2S MODE: **HALF-DUPLEX** SAMPLE-RATE: **44kHz** WORD-LENGTH: 32 I2S TEST: Master As Transmitter & Receiver As Slave With Write and Read Test	PASS	
1 - I2S write test		PASS	
2 - I2S write via interrupts test		PASS	
3 - I2S write via DMA test		PASS	
4 - I2S read test		PASS	
5 - I2S read async test		PASS	
6 - I2S read via DMA test		PASS	
7 - I2S Master Clock Disable		PASS	
0 - I2S initialization test	I2S MODE: **FULL-DUPLEX** SAMPLE-RATE: **96kHz** WORD-LENGTH: 32 I2S TEST: Master/Controller & Slave Target With Write and Read Test	PASS	
1 - I2S write test		PASS	
2 - I2S write via interrupts test		PASS	
3 - I2S write via DMA test		PASS	
4 - I2S read test		PASS	
5 - I2S read async test		PASS	
6 - I2S read via DMA test		PASS	
7 - I2S Master Clock Disable		PASS	

Peripherals Test Results – GPIO PIN MUX

GPIO Test Cases	Result	Remark
read gpio: **CMD: read 0 <0...7>**	PASS	
write gpio value: **CMD:write 0 1**	PASS	Tested LEDs on all pins from PD0 to PD7. The system becomes unresponsive during execution, requiring a board reset before proceeding to the next test. Requirement JIRA: USDK-294
gpioInt gpio: **CMD: gpioInt 0**	PASS	As per code no interrupt is enabled.
get_alt **: CMD: get_alt**	PASS	Printing default value for PORT
set_alt gpio value:**CMD: set_alt 0 7**	PASS	Tested For I2C,UART and SPI
pdl_test <test>: **CMD:pdl_test i2c**	PASS	I2C_0 & I2C_1 Tested only Master Mode using FRAM Sensor
pdl_test <test>: **CMD:pdl_test spi**	PASS	SPI_0 & SPI_1 Tested only Master Mode using FLASH Sensor
pdl_test <test>: **CMD:pdl_test uart**	PASS	UART_1 & UART_2 Tested Baud Rates
pdl_test <test>: **CMD:pdl_test i2s**	FAIL	GPIO PIN MUX for I2S is failing: USDK-295. I2S is working with default configuration using GPIO CLI.
pdl_test <test>: **CMD:pdl_test sdio**	BLOCKED	Blocked: HW team(Athiq) is preparing test board.

- DUT: Neon B0PP16
- SQA Test Report: NEON_B0_RC7.2_Result_9thApril2026.xlsx

Peripherals Test Results – Comparator and WDT

Comparator Test Cases	Sub-Module Test Selection	Result	Remark
0 - GPIO_0 test	0 - Comparator Test "Normal Mode" 1 - Comparator Test "ULP Mode" 2 - ESC	PASS	
1 - GPIO_1 test		PASS	
2 - GPIO_2 test		PASS	
3 - GPIO_3 test		PASS	
4 - GPIO_4 test		PASS	
5 - GPIO_5 test		PASS	
6 - GPIO_6 test		PASS	

WDT Test Cases	Result	Remark
0 - Test -> WDT IRQ (No reset)	PASS	
1 - Test -> WDT Reset	PASS	

Peripherals Test Results – uDMA and SDIO

uDMA Test Cases	Result	Remark
0 - uDMA Mem-to-Mem DMA transfer - Channel 0	PASS	
0 - uDMA Mem-to-Mem DMA transfer - Channel 1	PASS	
0 - uDMA Mem-to-Mem DMA transfer - Channel 2	PASS	
0 - uDMA Mem-to-Mem DMA transfer - Channel 3	PASS	
1 - uDMA SW triggered IRQ test - UDMA_DONE_CH0_IRQn	PASS	
1 - uDMA SW triggered IRQ test - UDMA_DONE_CH1_IRQn	PASS	
1 - uDMA SW triggered IRQ test - UDMA_DONE_CH2_IRQn	PASS	
1 - uDMA SW triggered IRQ test - UDMA_DONE_CH3_IRQn	PASS	
1 - uDMA SW triggered IRQ test - UDMA_ERR_IRQn	PASS	

SDIO Test Cases	Result	Remark
0 - SDIO device init	PASS	Tested with STM32H747 Host
1 - SDIO device-initiated host read transfer test	PASS	Tested with STM32H747 Host
2 - SDIO device send message to host test	PASS	Tested with STM32H747 Host
3 - SDIO de-init	PASS	Tested with STM32H747 Host

Peripherals Test Results – I2C

I2C-0 add I2C-1 Test Cases	Result	Remark
0 - I2C getVersion_Test	PASS	Executed using Automated test script and manual test.
1 - I2C Master write data IRQ mode test	PASS	Executed using Automated test script and manual test.
2 - I2C Master write data Polling mode test	PASS	Executed using Automated test script and manual test.
3 - I2C Master read data IRQ mode test	PASS	Executed using Automated test script and manual test.
4 - I2C Master read data Polling mode test	PASS	Executed using Automated test script and manual test.
5 - I2C slave receive data IRQ mode test	PASS	Executed using Automated test script and manual test.
6 - I2C slave receive data Polling mode test	PASS	Executed using Automated test script and manual test.
7 - I2C slave send data IRQ mode test	PASS	Executed using Automated test script and manual test.
8 - I2C slave send data Polling mode test	PASS	Executed using Automated test script and manual test.
9 - I2C Master send data DMA mode test	PASS	Executed using Automated test script and manual test.

Peripherals Test Results - UART

Test Cases (UART-1 & UART-2)	Result	Remark
0 - UART speed test (Baud rate)	PASS	3 UART TTL converters used: 1 for log and remaining 2 for UART Test and data verified.
1 - UART Data RX with Interrupts	PASS	Rx Data Size:1,8,16 Bytes – Working Rx Data Size: 32, 256,1024: Not Working
2 - UART Data RX into ring buffer	PASS	
3 - UART Data TX with interrupts	PASS	
4 - UART Test Break condition in LoopBack mode	PASS	
5 - UART DMA test RX	PASS	
6 - UART DMA test TX	PASS	30, 64,128, 256, 512,1024, 2048, 4096 bytes of data tested working.
7 - UART Loop Back Test	FAIL	Regression: Test application is crashing. JIRA USDK-288
8 - --- NOT SUPPORTED (UART IrDa)	BLOCKED	Test not enabled in SDK
9 - --- NOT SUPPORTED (UART RS-485)	BLOCKED	Test not enabled in SDK
a - UART RX with HW Flow Control	PASS	Rx Data Size:1,8,16 Bytes – Working Rx Data Size: 32, 256,1024: Not Working
b - UART TX with HW Flow Control	PASS	

Peripherals Test Results - Timers

Peripheral Module	Test Case	Result	Comments
LPTIMER	LPTIMER Display IP Version	PASS	
	LPTIMER Count Test	PASS	
	LPTIMER APB INT Test	PASS	
	LPTIMER TMR INT Test	PASS	
	LPTIMER SUSPEND RESUME Test	PASS	
DUAL-TIMER	Dual_Timer_1A test	PASS	
	Dual_Timer_1B test	PASS	
	Dual_Timer_2A test	PASS	
	Dual_Timer_2B test	PASS	
	Dual_Timer_3A test	PASS	
	Dual_Timer_3B test	PASS	
	Dual_Timer_4A test	PASS	
	Dual_Timer_4B test	PASS	
RTC	RTC getVersion_Test	PASS	
	RTC setDateTime_Test	PASS	
	RTC getCurrentTime_Test	PASS	
	RTC setAlarmDateTime_Test	PASS	
	RTC getAlarmDateTime_Test	PASS	
	RTC getCurrentTime_Test Loop	PASS	

Peripherals Test Results - xSPI

Test Case	Result	Remark
0 - xSPI init and get device ID test	PASS	
1 - xSPI clock frequency test	BLOCKED	Test not enabled in SDK
2 - xSPI sector erase, read and verify	PASS	
3 - xSPI chip erase, read and verify	PASS	
4 - xSPI flash read and write test - Polling mode	PASS	
5 - xSPI flash read and write test - Interrupt mode	BLOCKED	Test not enabled in SDK
6 - xSPI flash read and write test - DMA mode	BLOCKED	Test not enabled in SDK
7 - xSPI PSRAM read and write test - Polling mode	BLOCKED	Test not enabled in SDK
8 - xSPI PSRAM read and write test - Interrupt mode	BLOCKED	Test not enabled in SDK
9 - xSPI PSRAM read and write test - DMA mode	BLOCKED	Test not enabled in SDK
10 - xSPI PSRAM AIP read and write test	BLOCKED	Test not enabled in SDK
11 - xSPI switch to XIP mode	BLOCKED	Test not enabled in SDK
12 - xSPI switch to normal mode	BLOCKED	Test not enabled in SDK
13 - xSPI Single write, Single read in SDR mode	PASS	
14 - xSPI Single write, Dual read in SDR mode	PASS	
15 - xSPI Single write, Quad read in SDR mode	PASS	
16 - xSPI Quad write, Single read in SDR mode	PASS	
17 - xSPI Quad write, Dual read in SDR mode	PASS	
18 - xSPI Quad write, Quad read in SDR mode	PASS	

Critical Open Issues

Open issues - https://innophaseiot.atlassian.net/issues/?filter=13347

JIRA	Issue	Comments
USDK-7	Data stalls and disconnections observed during TCP DL traffic with TP-Link AX6000, Aruba 534 and Xiaomi AX6000	
USDK-6	Data stalls observed during TCP UL traffic with Xiaomi AX6000 and Aruba 534	
USDK-83	UDP DL peak tput is very low with 90M UDP BW	Better with 50M
USDK-16	STA disconnects due to ‘Out of memory’ during TCP traffic tests	
USDK-200	Data stalls observed during TCP traffic in noisy environment	
USDK-203	DL TxBF-OFF Peak throughput is lower (30% lower) when compared to TxBF-ON	
USDK-141	RvR near range tput is lower for all traffic types	Further reduced for DL traffic.
USDK-21	Memory leak observed during TCP DL traffic at lower RSSI (-70 dBm)	
USDK-85	BLE connection is unstable during tput and gatt operations with all PHYs	
USDK-79	Disconnects observed with IFX devkit.	
USDK-243	Variations in UL tput are seen across NeonB0 boards	
USDK-238	Beacon reception time for DTIM 100 (~10 ms) is higher compared to lower DTIM Values (~3 ms)	
USDK-239	High current spike during beacon wake for high DTIM (~1.2 A for DTIM100 Vs ~800–900 mA for DTIM1/DTIM2)	
USDK-53	Hwaddr, suspend, ssid and passphrase boot args do not take effect.	
USDK-245	Soft SWD reset does not work on INP3119 EVK and INP6120 RK EVBs	

Backup

Jan 30, 2026

Challenges and Path forward

Technical

1. **Flash Architecture:** Rearchitected Flash boot design to resolve NPU loading issues caused by Neon.B0 partitioning differences
    1. Thanks to Per's guidance and great collaboration effort between Sweden, China (Mason and team), and SJ team (Yuhui, Zhibin)

2. **Power Management:** Enabled APU deep sleep and NPU suspend within 2 days **(Thanks to Per and Ander's guidance)**
    1. Another great collaboration effort between Sweden, China (Haoming, Mason) and San Diego (Vova, Dan) teams
    2. Demo app is included in this release

3. **Hardware Selection:** Identified **"goldenish" board** to bypass RF instability and board-to-board variation
    1. Observed both in WiFi, BLE bring-up/debug support
    2. 3-4 weeks effort spent on debugging/identifying/WAR. Thanks to Kevin and SJ team (Bill, Zhibin, Yuhui) for a path forward

4. **Stability WAR:** Disabled beamforming/sounding to mitigate iperf instability when both Sounding/PS is enabled
    1. Zhibin, Mohammad, Dan and Anders are collaborating to debug the root-cause of stability issues seen during iperf testing

5. **BLE connectivity:** Narrowed down Neon.B0 BLE Central showing instability/connectivity issues
    1. Validation underway with vcm fixes for PER hump, potentially address 10dB+ degradation in the RF wedge
    2. Further clarity on Neon.B0 RF settings/cal is needed to root-cause the BLE stability/connection issues
    3. Recommend to use Neon.B0 as Peripheral with this release (Pankaj, Sunil)

Challenges and Path forward

Processes

1. **Stability:** Branch cut-off (Dec-22) and targeted NPU fixes merged to narrow down stability issues (Yuvaraj)
    1. Multiple functionality across NPU is in various stages of implementation, causing instability/regression on NPU main

2. **Resource Management:** Streamlined and prioritized requirements/resources
    1. Support FPGA verification for Helium.B0, IFX and Peripheral support on Neon.B0 bring-up in parallel

3. **Documentation challenges:** Multiple design documents meetings held for clarity
    1. In house designed components (e.g. SysPM) needs better documentation requiring more engineering time for development and debugging

4. **Management Asks:** Clarity on requirements, timelines and deliverables from management

5. **Testing Asks:** Unified Test platforms, standardized/unified HW test components, comprehensive test plans

WiFi Feature status

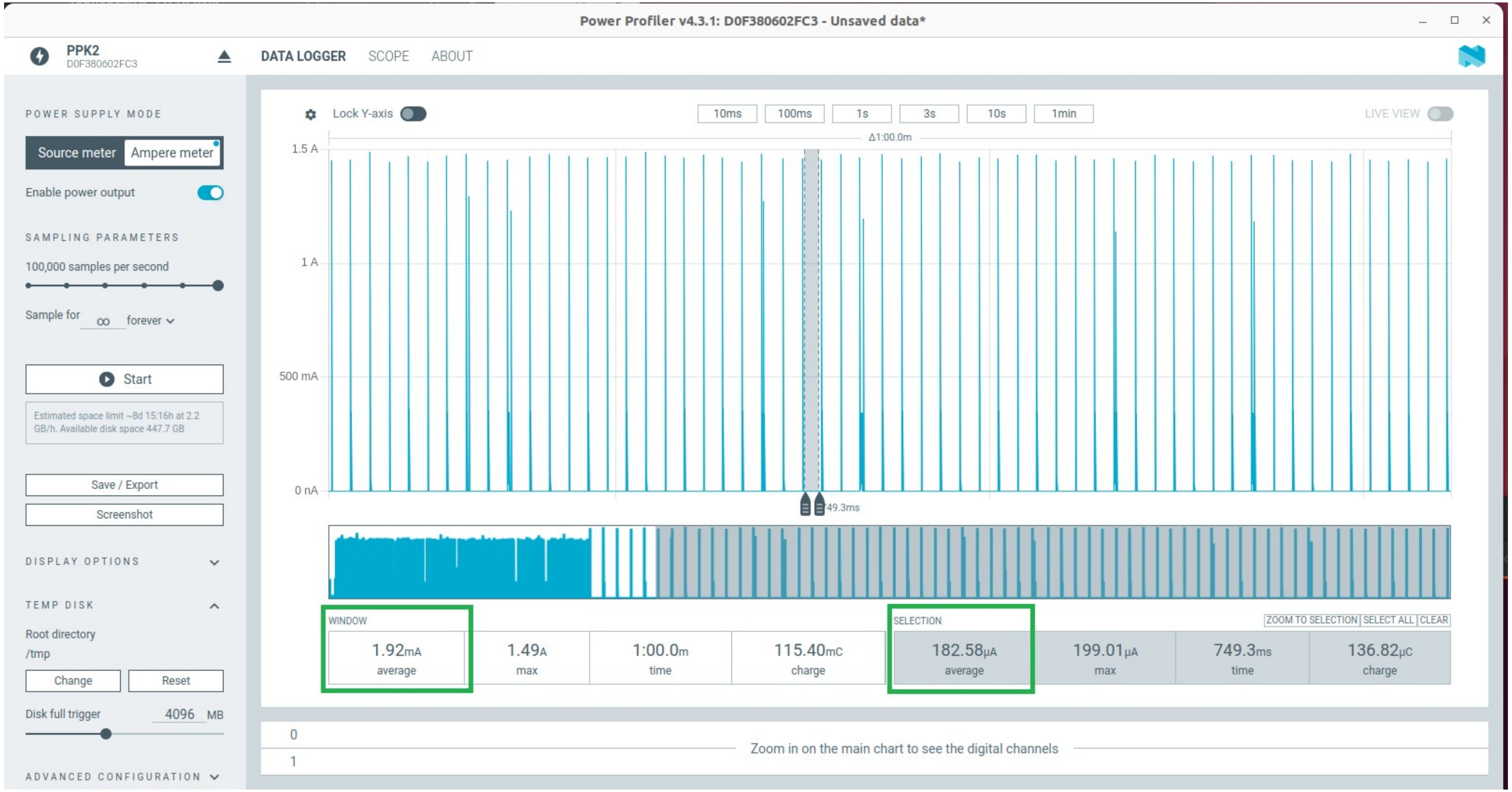
Feature	SW status	Owner	Comments
MCS 8/9 for 11ax		Anders	SW status: Supported
BSS Coloring		Yuhui	SW status: Supported
WPA2/WPA3 PSK/CCMP		Bill, Vamsi	SW status: Supported
Protocol Power Save		Anders, Zhibin, Bill	SW status: Supported
Low Power (Suspend)		Anders, Vova, Haoming, Yuhui	Basic functionality supported in APU and NPU
2nd stage bootloader		Per, Mason, Yuhui, Zhibin	SW status: Supported
Sounding response, BFmee		Patrik, Jigang	Disabled by default due to stability issues

BLE Feature status

Feature	SW status	Owner	Comments
Simultaneous BLE		Patrik, Pankaj, Sunil	SW status: Supported
Periodic Advertisements		Patrik, Pankaj, Sunil	SW status: Supported
LE connections		Patrik, Pankaj, Sunil	SW status: Supported
Extended scan		Patrik, Pankaj, Sunil	SW status: Supported
Adaptive Channel Hoppint		Patrik, Pankaj	SW status: Supported

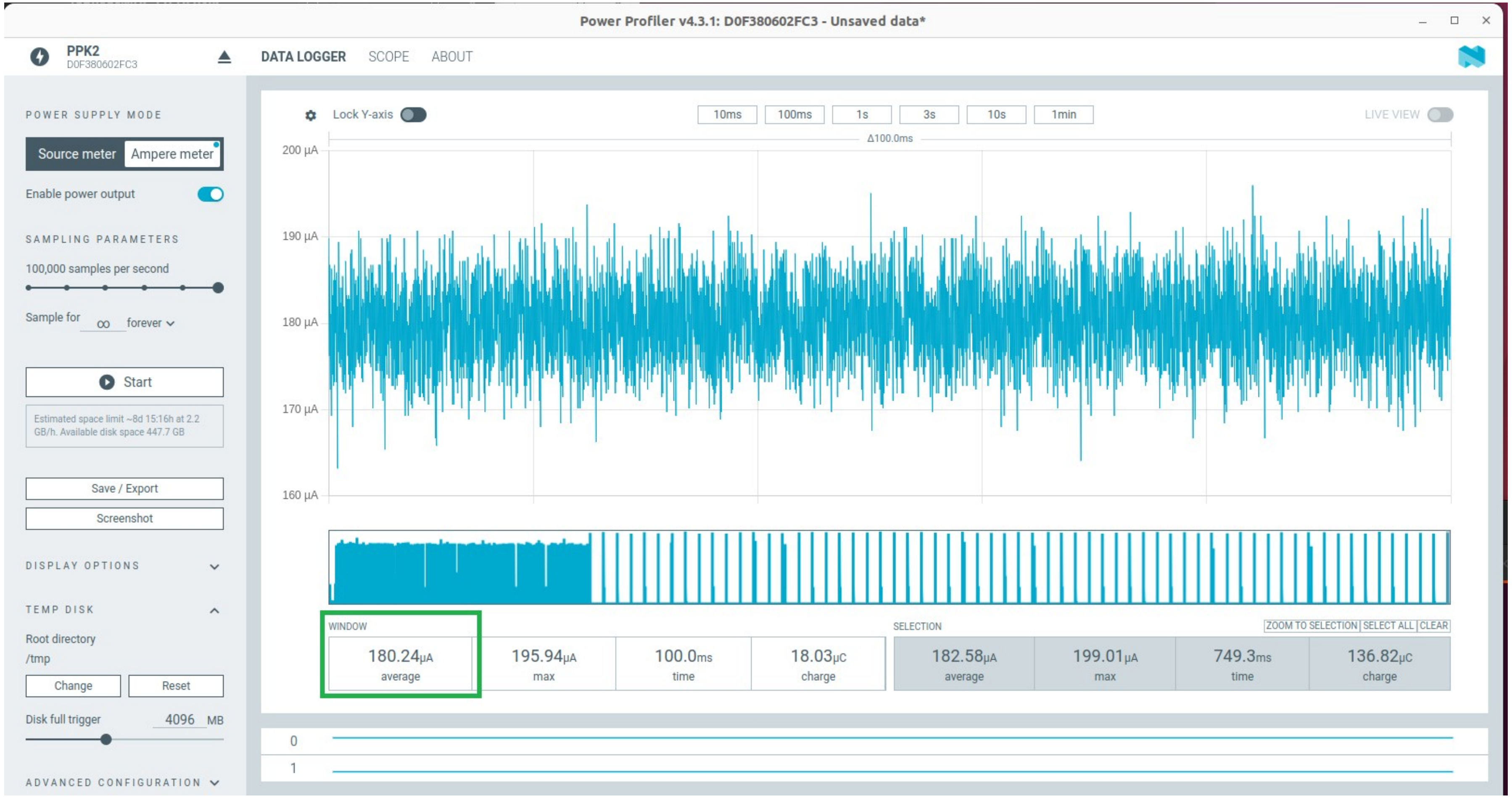
Low Power current consumption

The DTIM10 average current is about 1.92mA and the suspend current is around 180.24uA



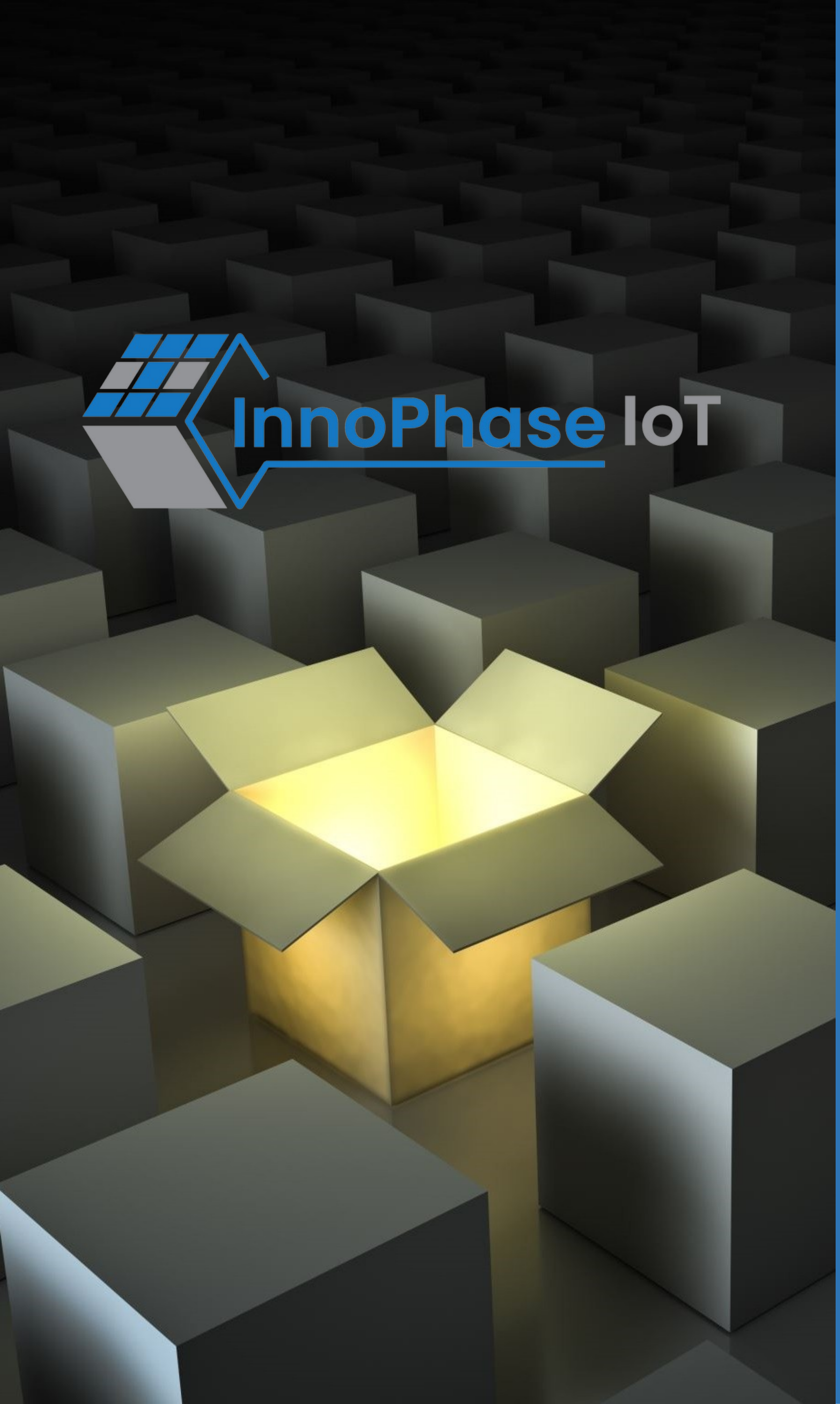
Low Power current consumption

The DTIM10 average current is about 1.92mA and the suspend current is around 180.24uA



Peripheral status

Peripheral	SW Status	Owner	Comments
SPI		Vova	SW status: PDL Driver supported *SysPM driver - WIP
I2C		Vova	
RTC		Vova	
WDT		Vova	
UART		Vova	
*SysPM		Vova	
Timers		Vova	
XSPI_1, Flash XPSI_2, PSRAM		Shalini	
GPIO		Shalini	
SDIO		Shalini	



SW Tenets

SW Team Tenets

SW team Tenet

1. Simple, Stable, Scalable, Usable, Functional code
2. Unified team, Unified code
3. Continuous development, Continuous Delivery

Thanks for the great collaboration effort across Sweden, China, India, San Diego and San Jose teams

WiFi EVM Test

TX_EVM_TEST			Neon B0 33# 20260108									Neon B0 29# 20251126									20260108 Avg	20251126 Avg
			test1			test2			test3			test1			test2			test3				
Protocol	Modulation	Baseline	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462		
802.11b	DSSS-1	-10	-23.11	-23.27	-23.33	-22.84	-15.22	-15.49	-22.92	-23.27	-23.02	-23.13	-23.53	-23.48	-23.68	-22.92	-23.81	-23.59	-24.12	-23.63	-21.39	-23.54
	DSSS-2	-10	-22.61	-23.48	-23.40	-22.14	-23.25	-23.53	-23.33	-23.27	-22.70	-23.66	-23.69	-23.32	-23.51	-23.59	-23.48	-22.94	-23.34	-23.92	-23.08	-23.49
	CCK-5.5	-10	-24.03	-24.33	-24.20	-24.56	-24.64	-23.61	-23.52	-23.50	-24.14	-23.79	-23.69	-23.59	-23.62	-23.64	-23.71	-23.90	-24.94	-23.82	-24.06	-23.86
	CCK-11	-10	-23.59	-23.93	-23.74	-23.88	-23.02	-23.81	-23.57	-22.93	-23.42	-23.42	-23.53	-23.76	-23.51	-23.46	-23.06	-24.19	-23.22	-23.29	-23.54	-23.49
802.11g	OFDM-6	-5	-15.30	-15.21	-15.17	-15.41	-14.81	-14.29	-15.06	-15.46	-15.52	-15.54	-15.66	-14.87	-15.64	-15.78	-15.44	-14.80	-15.30	-15.23	-15.14	-15.36
	OFDM-9	-8	-16.28	-17.45	-17.24	-16.90	-16.07	-15.09	-17.08	-17.76	-18.34	-17.31	-17.26	-17.86	-17.04	-16.96	-16.86	-16.16	-17.00	-16.76	-16.91	-17.02
	OFDM-12	-10	-18.65	-18.00	-19.56	-18.30	-17.49	-16.93	-17.59	-17.72	-18.92	-19.19	-18.98	-18.91	-19.05	-19.21	-17.36	-18.28	-17.51	-18.78	-18.13	-18.58
	OFDM-18	-13	-20.41	-19.09	-19.69	-19.08	-18.09	-18.11	-20.49	-17.92	-19.69	-20.94	-19.32	-18.43	-20.03	-21.52	-19.88	-20.79	-19.90	-19.71	-19.17	-20.06
	OFDM-24	-16	-20.75	-20.72	-21.84	-20.30	-19.77	-20.32	-21.75	-20.98	-18.31	-21.45	-19.47	-22.23	-22.16	-22.26	-18.73	-20.77	-20.68	-21.43	-20.53	-21.02
	OFDM-36	-19	-20.88	-20.86	-23.34	-20.33	-20.22	-19.98	-22.69	-21.91	-20.96	-21.73	-21.38	-21.40	-21.86	-23.22	-21.94	-21.89	-21.03	-21.07	-21.24	-21.73
	OFDM-48	-22	-22.20	-23.13	-19.99	-22.72	-20.14	-18.45	-22.07	-22.84	-21.82	-23.34	-22.47	-23.43	-23.71	-22.91	-23.18	-23.15	-22.68	-21.82	-21.48	-22.97
	OFDM-54	-25	-22.81	-22.86	-21.01	-22.67	-21.82	-22.06	-22.53	-23.18	-24.44	-23.75	-21.53	-22.84	-24.24	-23.66	-20.61	-23.47	-22.95	-21.04	-22.60	-22.68
802.11n	MCS0	-5	-15.04	-14.71	-14.90	-14.78	NULL	NULL	-15.60	-14.21	-15.71	-14.39	-15.28	-15.09	-15.66	-16.16	-15.60	-14.49	-14.72	-15.49	-14.99	-15.21
	MCS1	-10	-16.33	-15.93	NULL	-16.09	NULL	-15.99	-16.61	NULL	NULL	-16.80	-16.71	-16.38	-16.18	-17.39	-16.65	-16.36	-16.45	-16.21	-16.19	-16.57
	MCS2	-13	-16.43	-17.88	-18.01	NULL	-16.65	-17.06	NULL	-17.86	-18.54	-17.59	-17.88	-17.77	-18.36	-18.42	-17.78	-18.15	-18.45	-17.33	-17.49	-17.97
	MCS3	-16	-19.25	-19.04	-19.44	-18.96	-18.61	-18.67	NULL	-20.53	-20.62	-19.88	NULL	-19.24	-20.34	-20.43	-19.09	-20.10	-18.88	-19.71	-19.39	-19.71
	MCS4	-19	-20.70	NULL	-18.93	-20.88	NULL	-18.29	NULL	-20.01	NULL	-21.44	-20.65	-19.83	-21.80	-22.61	NULL	-21.73	-21.33	-21.57	-19.76	-21.37
	MCS5	-22	-23.00	-21.74	-21.38	-21.21	-19.67	-21.26	-20.99	-22.27	-19.83	-23.56	-21.33	-21.99	-23.96	-23.15	-21.60	-23.10	-22.36	-20.60	-21.26	-22.40
	MCS6	-25	-22.39	-21.21	NULL	NULL	-19.31	-20.09	-20.81	-21.43	-21.01	-24.72	-23.61	-22.36	-22.94	-21.85	NULL	-23.18	-21.72	-21.11	-20.89	-22.69
	MCS7	-27	NULL	-24.24	-22.25	-20.88	-20.92	-20.77	-22.29	-22.87	-20.89	-23.66	-22.62	-23.18	-23.68	-21.22	-21.53	-23.27	-22.45	-21.77	-21.89	-22.60
802.11ax	MCS0	-5	-12.61	-13.10	-12.88	-14.12	-13.63	-13.17	-13.13	-14.01	-13.97	-15.78	-12.72	-16.49	-16.45	-14.02	-13.07	-16.07	-15.72	-14.99	-13.40	-15.03
	MCS1	-10	-13.83	-13.97	-17.31	-17.05	-13.32	-17.77	-14.66	-15.15	-16.87	-18.10	-19.07	-12.81	-18.55	-17.39	-17.01	-18.37	-16.98	-18.34	-15.55	-17.40
	MCS2	-13	-14.15	-15.58	-19.15	-17.10	-18.11	-12.56	-18.45	-14.74	-13.73	-19.33	-18.35	-15.12	-13.64	-18.50	-18.98	-19.54	-19.51	-19.87	-15.95	-18.09
	MCS3	-16	-15.52	-14.10	-15.24	-16.90	-15.00	-14.09	-15.66	-14.86	-21.68	-20.01	-15.54	-12.37	-15.34	-19.41	-14.44	-20.38	-19.87	-20.67	-15.89	-17.56
	MCS4	-19	-15.14	-13.94	-12.73	-21.75	-21.21	-18.18	-15.01	-15.65	-12.86	-21.05	-22.46	-20.72	-15.12	-23.25	-22.88	-20.27	-20.46	-15.08	-16.27	-20.14
	MCS5	-22	-17.24	-17.26	-16.98	-21.25	-17.60	-17.28	-21.66	-17.28	-20.42	-22.73	-23.00	-22.75	-16.84	-23.79	-16.49	-23.72	-19.90	-22.13	-18.55	-21.26
	MCS6	-25	-17.45	-17.22	-17.49	-20.45	-17.83	-17.31	-17.43	-17.62	-17.77	-19.62	-16.49	-16.39	-23.89	-16.85	-17.14	-15.95	-19.75	-18.29	-17.84	-18.26
	MCS7	-29	-17.65	-16.96	-19.82	-21.95	-17.08	-18.74	-21.45	-17.92	-17.10	-19.38	-17.58	-15.92	-16.30	-23.80	-16.60	-18.55	-21.95	-17.11	-18.74	-18.58
	MCS8	-32	-21.02	-22.34	-21.20	-22.18	-22.19	-22.13	-28.18	-21.90	-21.23	-22.19	-22.29	-22.46	-22.51	-22.17	-26.76	-22.81	-22.37	-22.33	-22.49	-22.88
	MCS9	-35	-22.84	-20.70	-22.32	-22.94	-25.22	-21.47	-22.33	-22.21	-22.73	-27.90	-22.41	-21.75	-22.09	-22.21	-26.31	-22.24	-22.29	-22.76	-22.53	-23.33

Neon SQA Test Status – USDK 1.1.1-EA1-RC5

Jan 15, 2025

InnoPhase IoT

Wi-Fi STA Peak Tput

WPA2-PSK	UDP DL	UDP UL	TCP DL	TCP UL
11ax	48.1	31.5	32.8	26.8
11n	48.3	29.3	28.8	25.4
11g	27.1	18.7	14.2	10.1
11b	7.7	7.0	6.2	6.2

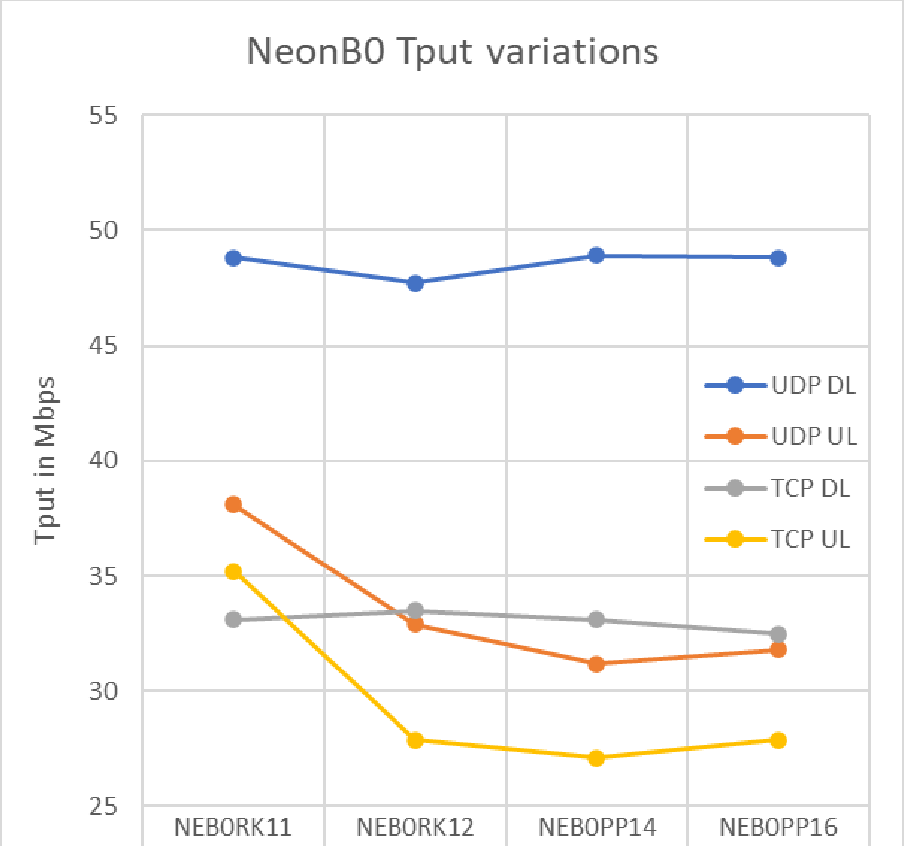
- DUT: Neon B0PP14; Tput in Mbps.
- Build: USDK 1.1.1-EA1-RC5 (NPU: git-96b31d55; APU: 86a936b8)
- Test AP: Asus RT-AX86U PRO; RSSI: -31 dBm
- Test Setup: Cabled inside RF Shield box; Clean channel

Wi-Fi STA Peak Tput – OTA Vs Cabled

Test Environment	RSSI	UDP DL	UDP UL	TCP DL	TCP UL
11ax – Cabled	-30 dBm	48.1	31.5	32.8	26.8
11ax – OTA	-26 dBm	46.9	30.8	31.1	26.1
Delta (OTA vs Cabled)		-2.6%	-2.3%	-5.5%	-2.7%

- DUT: Neon B0PP14; Tput in Mbps.
- Test AP: Asus RT-AX86U PRO
- Cabled Test setup: Inside RF Shield box; clean channel – CH6
- OTA Test setup: Inside Anechoic chamber; clean channel – CH6

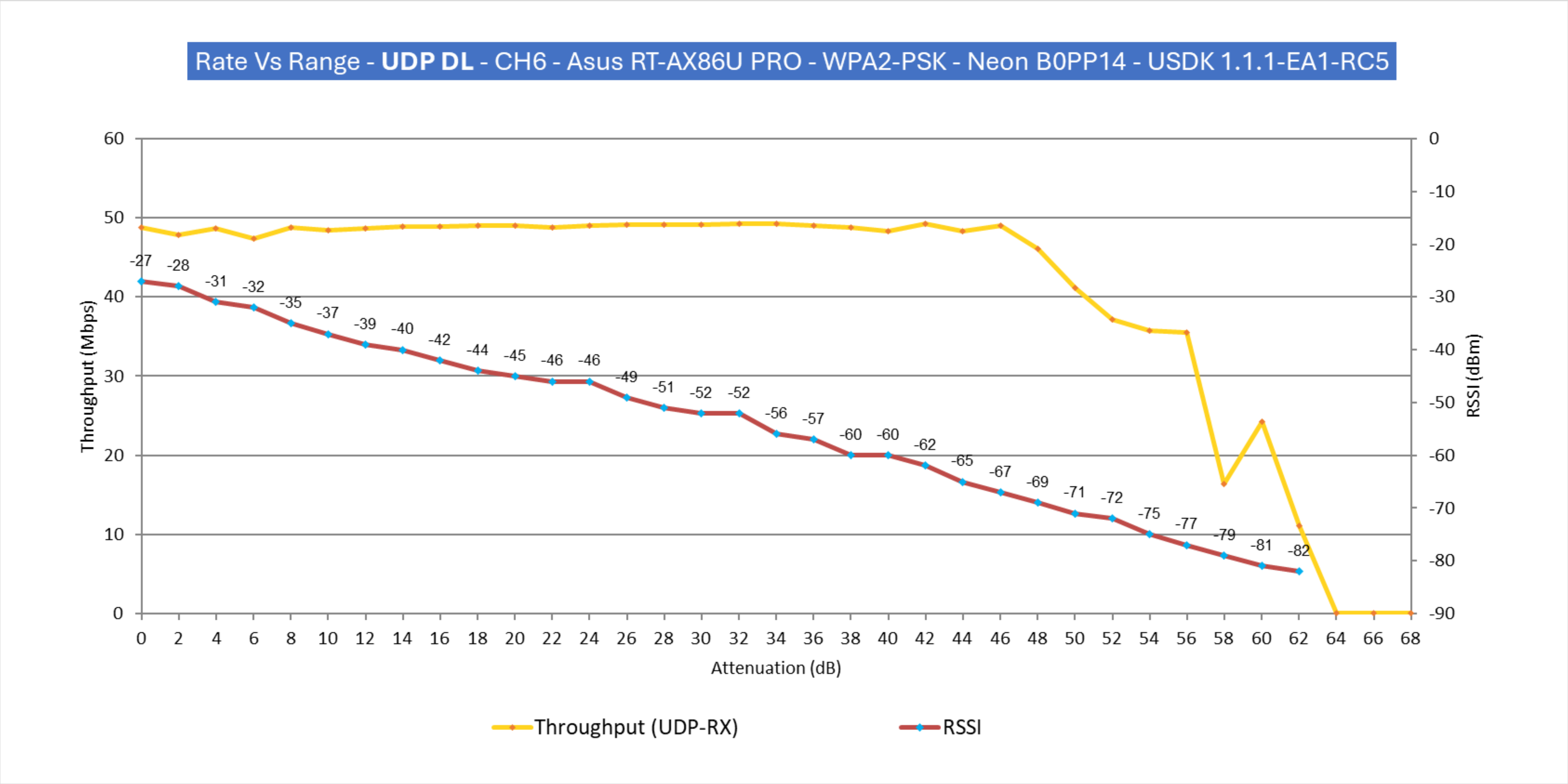
Wi-Fi STA Peak Tput – Board variations



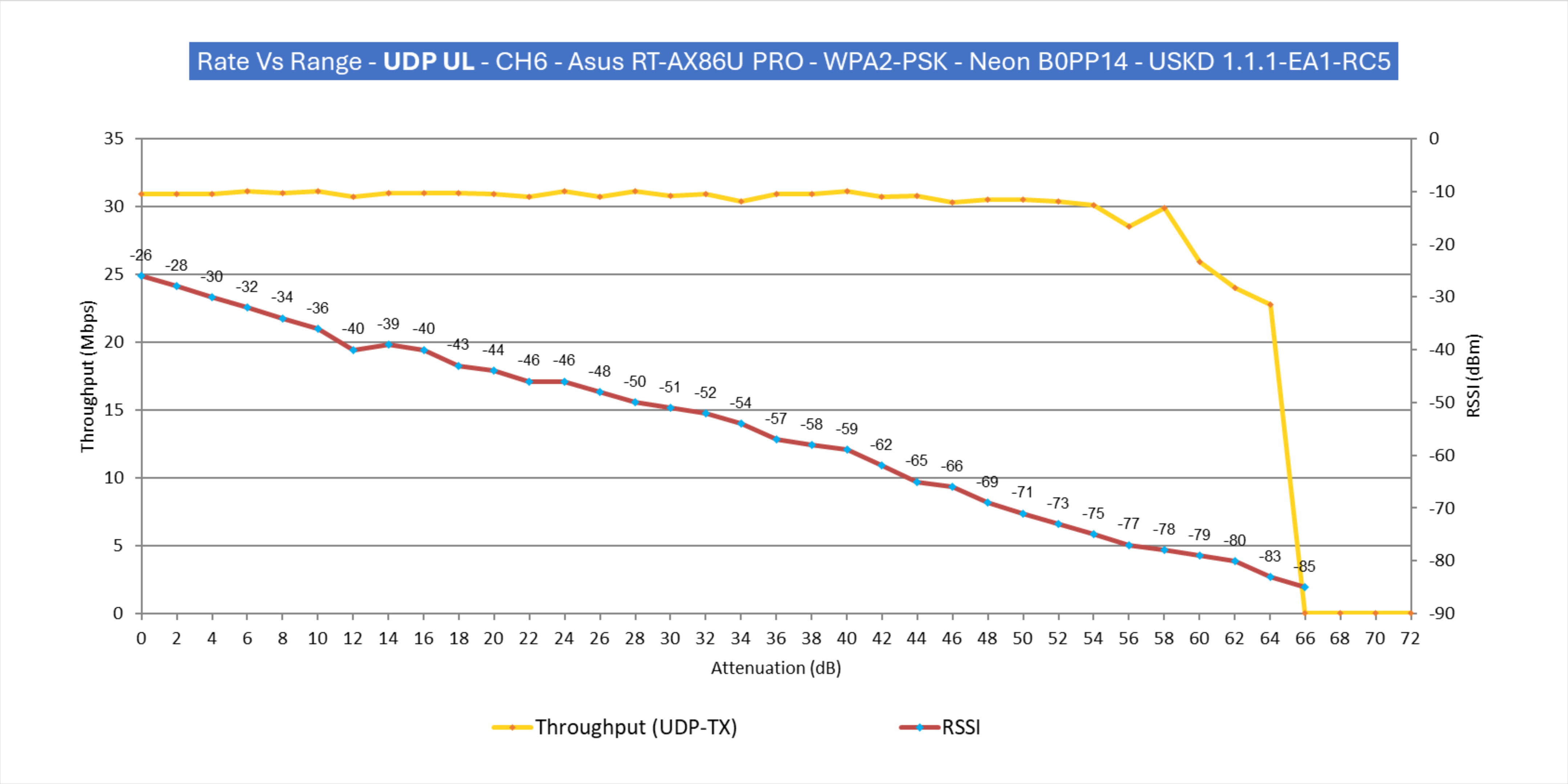
	NEB0RK11	NEB0RK12	NEB0PP14	NEB0PP16
UDP DL	48.8	47.7	48.9	48.8
UDP UL	38.1	32.9	31.2	31.8
TCP DL	33.1	33.5	33.1	32.5
TCP UL	35.2	27.9	27.1	27.9

- Test AP: Asus RT-AX86U PRO
- Test Setup: Cabled inside RF Shield box; Clean channel
- RSSI: around -30 dBm
- NEB0RK – EVB (small board)
- NEB0PP – EVK (production package)

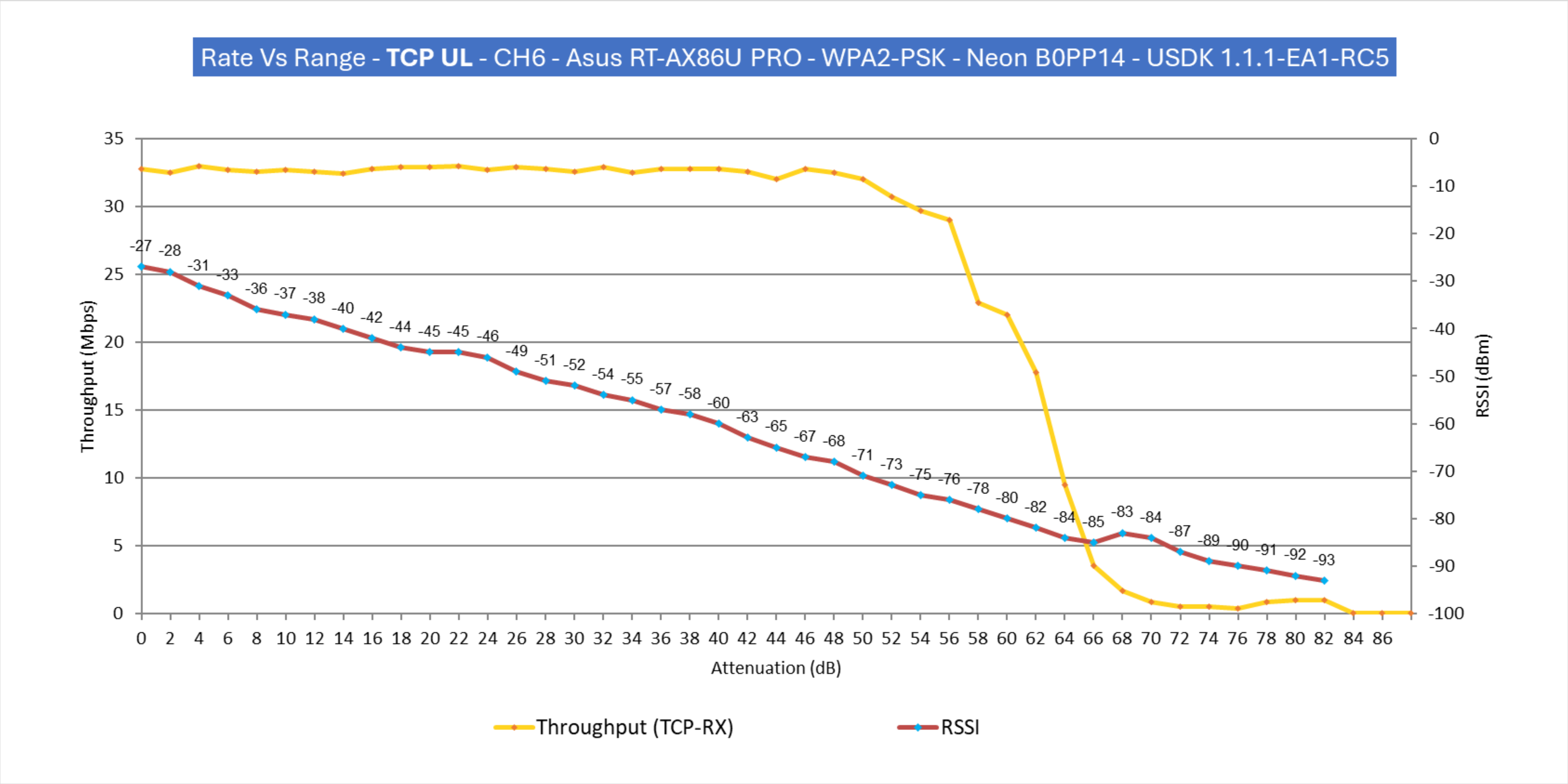
RvR – UDP DL – WPA2-PSK



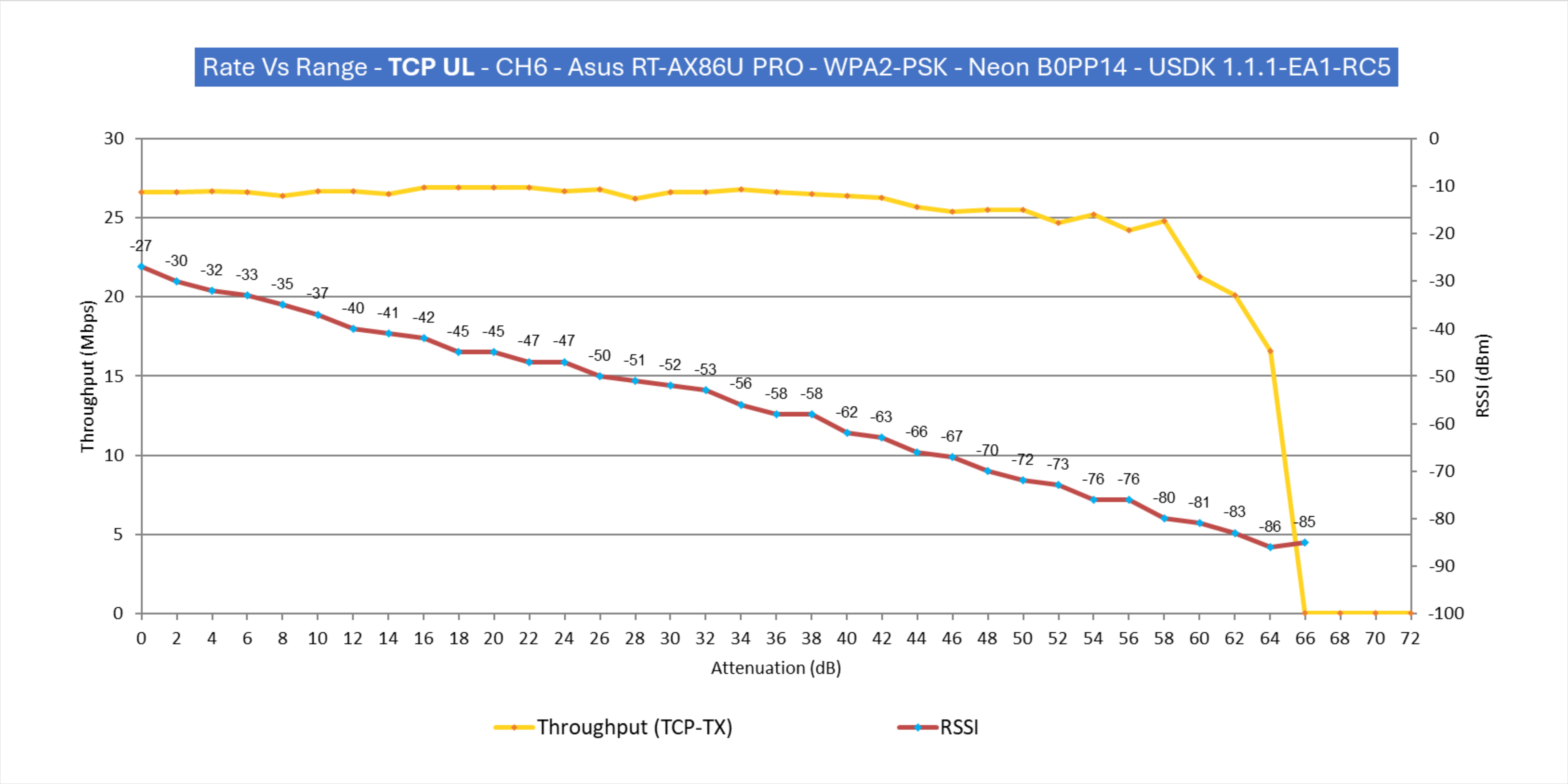
RvR – UDP UL – WPA2-PSK



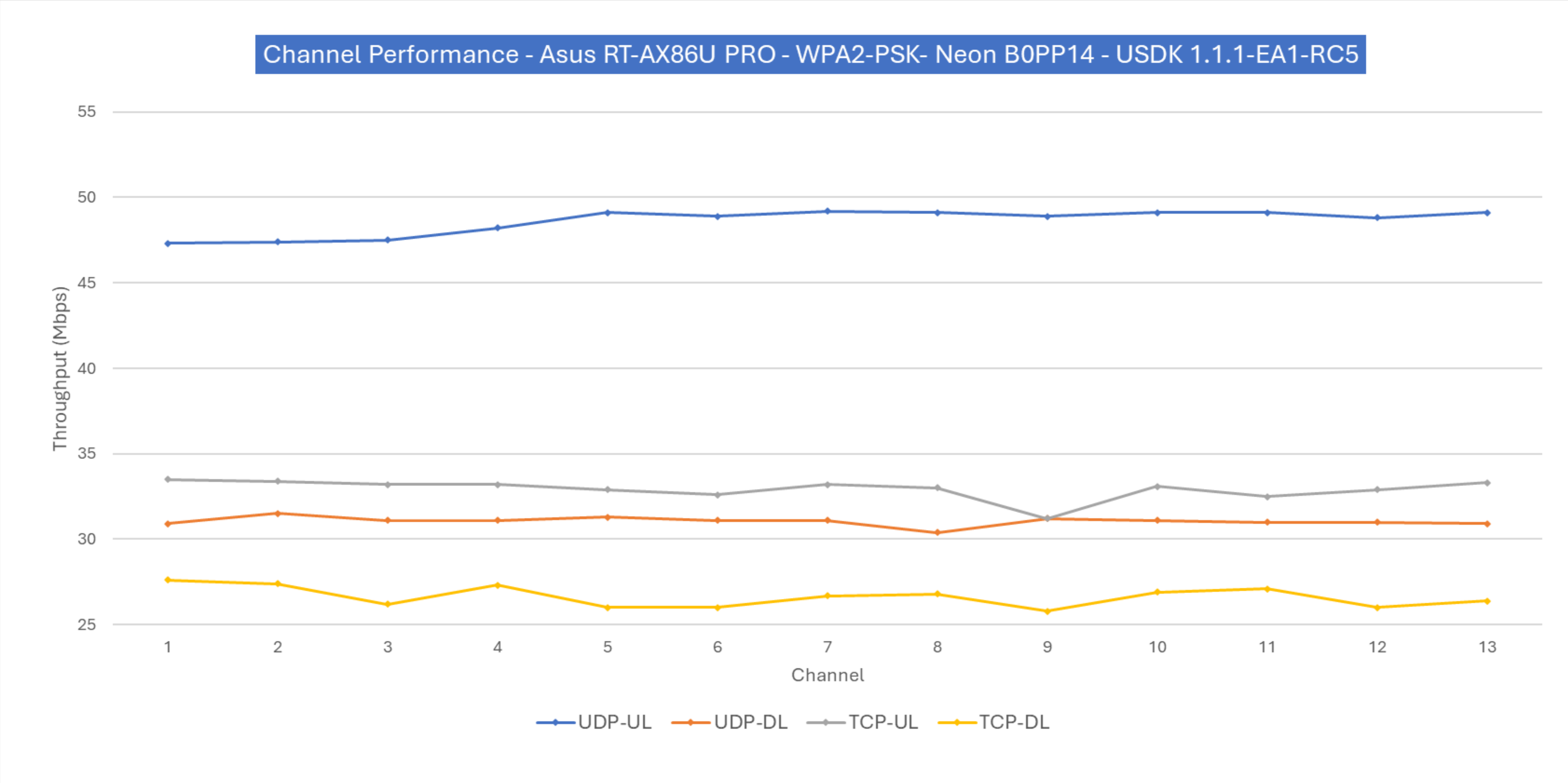
RvR – TCP DL – WPA2-PSK



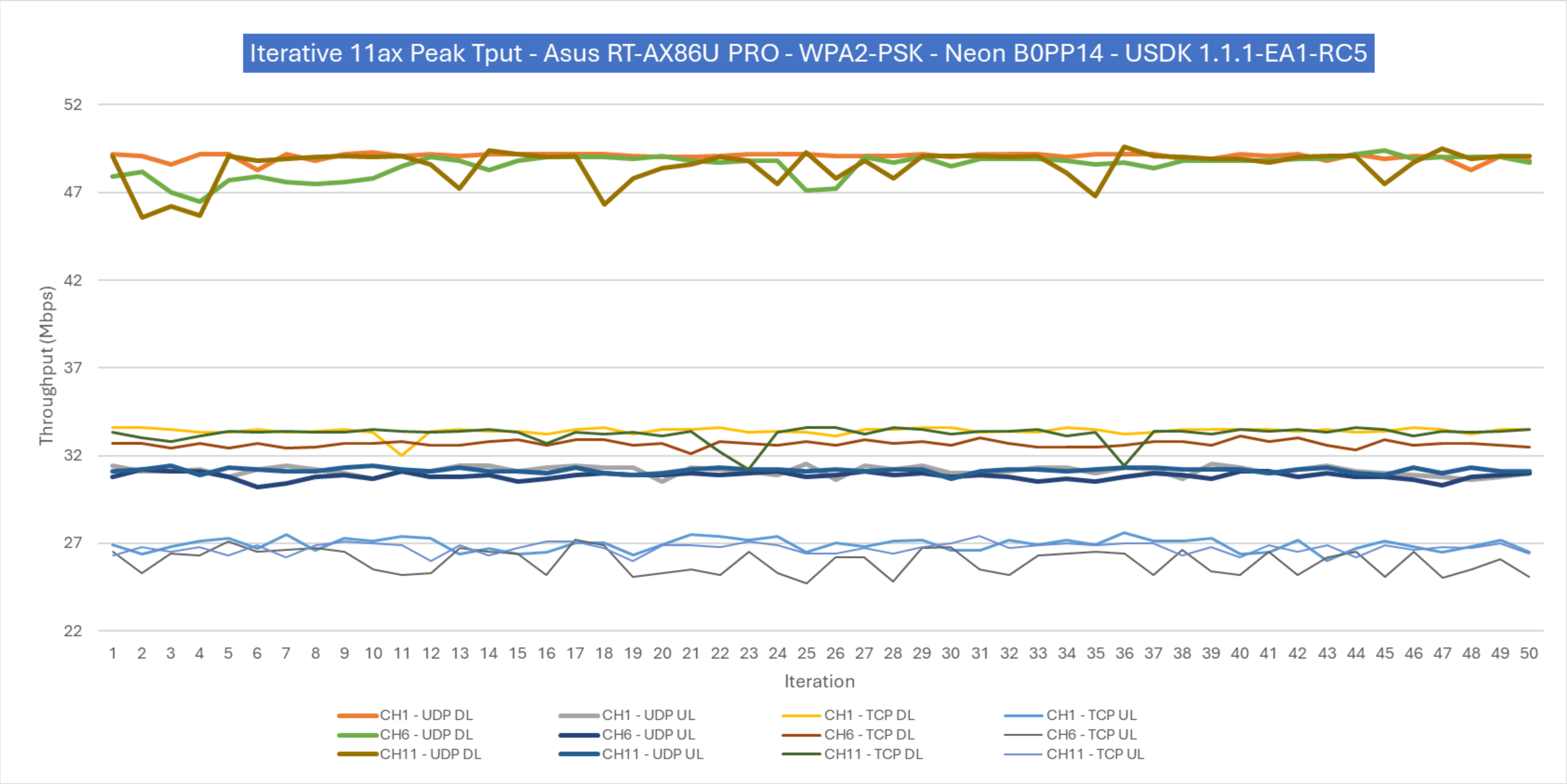
RvR – TCP UL – WPA2-PSK



Channel Performance – WPA2-PSK



Repeated Tput – WPA2-PSK



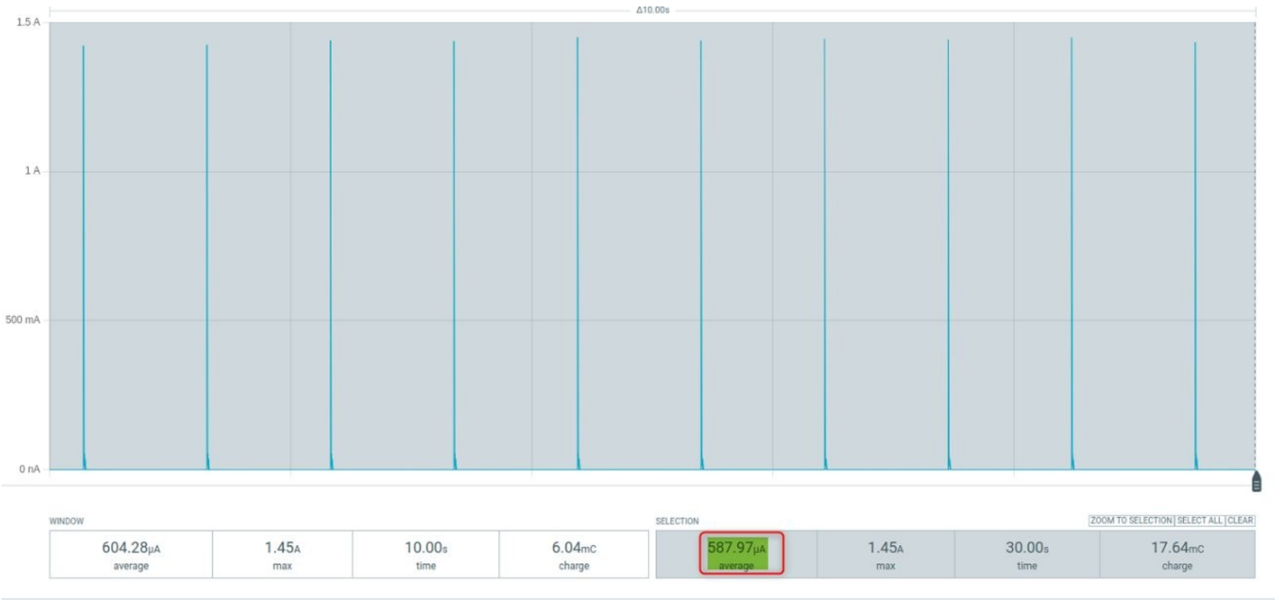
Wi-Fi STA Power save

DTIM Interval	Functionality	Avg Current (mA)	Comments
DTIM1	PASS	28.6	• DTIM functionality tested working using power profiler. Wakeups align with DTIM set on the AP. • QoS Null for sleep-wakeup verified using sniffer. • Avg current is higher due to device not entering suspend with test apps – wifi_connect, power_save.
DTIM3	PASS	27.6	
DTIM5	PASS	27.5	
DTIM10	PASS	27.3	
DTIM30	PASS	27.2	
DTIM100	PASS	27.1	

- DUT: Neon B0PP10
- Test AP: Asus RT-AX86U PRO; RSSI: -34 dBm
- Test Setup: Cabled inside RF Shield box; Clean channel

Wi-Fi STA Low Power – RC5

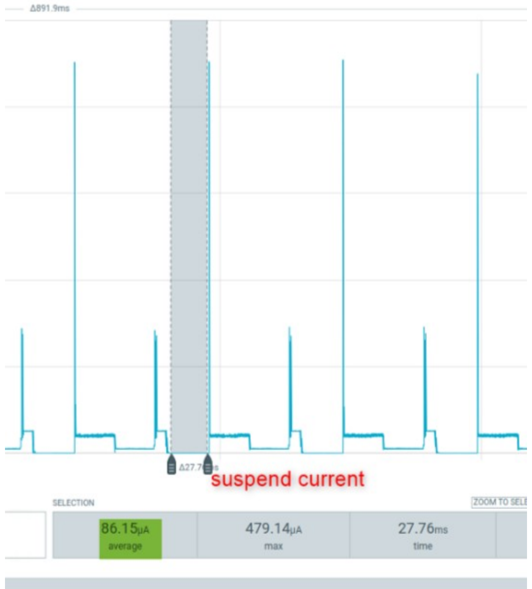
- Neon B0 entering suspend has been tested working using udp_demo application.
- **Suspend current** measured on NEB0PP10 is around **75uA**
- Average **DTIM10 current** measured using udp_demo app is around **587uA**

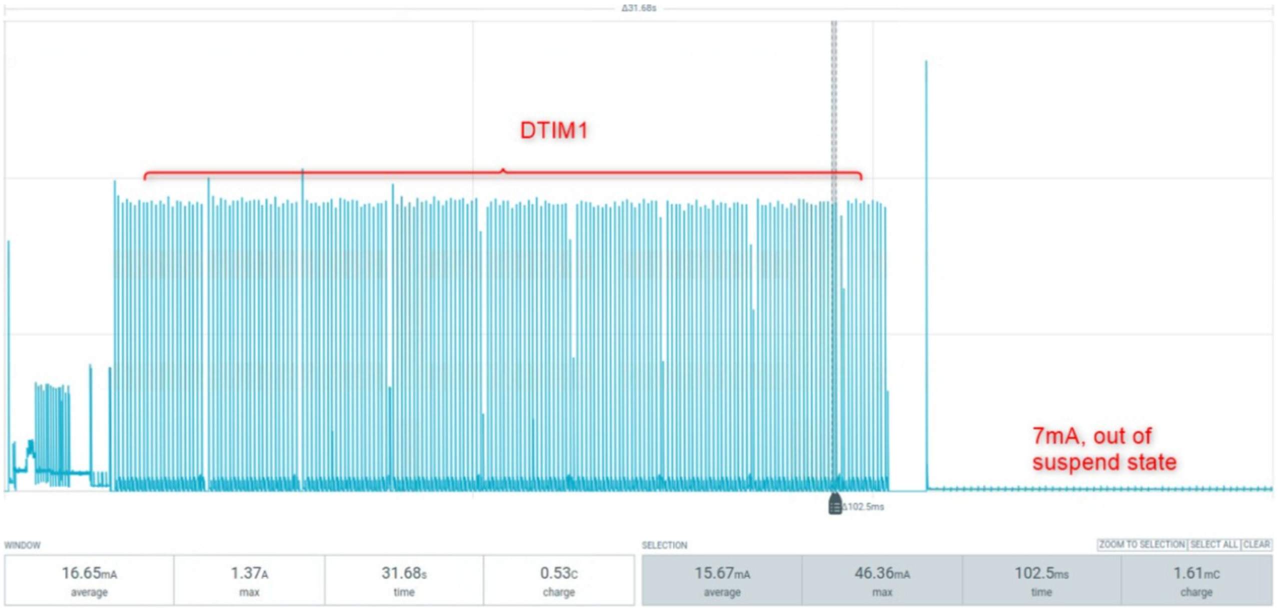


- DUT: Neon B0PP10
- Test AP: Asus RT-AX86U PRO; RSSI: -34 dBm
- Test Setup: Cabled inside RF Shield box; Clean channel

Wi-Fi STA Low Power – RC5p3

- Neon B0 entering suspend has been tested working using udp_demo_rc5p3 application.
- **Suspend current** measured on NEB0PP10 board is around **86uA**
- Average **DTIM1 current** measured using udp_demo_rc5p3 app is around **15mA**. DUT does not enter DTIM10 and exits low power state in ~20secs





- DUT: Neon B0PP10
- Test AP: Asus RT-AX86U PRO; RSSI: -34 dBm
- Test Setup: Cabled inside RF Shield box; Clean channel

Wi-Fi STA Interoperability – WPA2-PSK

Test AP	Wi-Fi Chipset	RSSI	UDP DL	UDP UL	TCP DL	TCP UL
ASUS RT-AX86U Pro	Broadcom BCM43684	-32 dBm	41	31	29	27
TP-Link AX6000	Broadcom BCM43684	-34 dBm	18	0	0	0
ASUS RT-AX88U	Broadcom BCM43684	-35 dBm	2.5	22	0	0
Mercusys MR80X	MediaTek MT7981BA	-36 dBm	4 (46)	34	0	0
Huawei WS7200 AX3	Hisilicon (Huawei) Hi1152	-40 dBm	0.3	35	18	28
Xiaomi MI RA72 AX6000	Qualcomm IPQ5018	-44 dBm	44	35	0	0
TP-Link BE5100	MediaTek MT7991AV	-44 dBm	22 (34)	33	20	0
TP-Link wifi-7 BE6500	Broadcom BCM6766	-33 dBm	43	30	24	24
Aruba-534	Qualcomm	-34 dBm	23 (37)	36	0	0

- DUT: Neon B0PP16. Tput in Mbps.
- Test Setup: Radiated inside RF Shield box; Clean channel
- UDP BW: 90M (50M)

Wi-Fi STA Stability

Test Scenario	Test duration	Channel	Test AP	Result	Comments
TCP DL	2 Hours	**Clean**	Asus RT-AX86U PRO	PASS	Tput at end of test – 33.5 Mbps
TCP UL	2 Hours	**Clean**	Asus RT-AX86U PRO	FAIL	WD timeout observed in 95 mins
UDP DL	2 Hours	**Clean**	Asus RT-AX86U PRO	PASS	Tput at end of test – 47.5 Mbps
UDP UL	2 Hours	**Clean**	Asus RT-AX86U PRO	PASS	Tput at end of test – 37 Mbps
Ping traffic - 32B - DTIM3	12 Hours	**Clean**	Asus RT-AX86U PRO	PASS	
Ping traffic - 32B - DTIM10	18 Hours	**Clean**	Asus RT-AX86U PRO	PASS	
TCP DL	2 Hours	**Noisy**	Asus RT-AX86U PRO	FAIL	WD timeout observed in 2 mins
TCP UL	2 Hours	**Noisy**	Asus RT-AX86U PRO	PASS	Tput at end of test – 5.3 Mbps
UDP DL	2 Hours	**Noisy**	Asus RT-AX86U PRO	PASS	Tput at end of test – 5.2 Mbps
UDP UL	2 Hours	**Noisy**	Asus RT-AX86U PRO	PASS	Tput at end of test – 6.7 Mbps
TCP DL	2 Hours	**Noisy**	D-Link X6060	FAIL	WD timeout observed in 17 mins
TCP UL	2 Hours	**Noisy**	D-Link X6060	FAIL	Disconnect (Deauth7) in 5 mins
UDP DL	2 Hours	**Noisy**	D-Link X6060	FAIL	WD timeout observed in 25 mins
UDP UL	3 Hours	**Noisy**	TP-Link AXE7800	FAIL	WD timeout observed in 2 hours

Noisy channel condition is case-3 (high interference) environment; BLR office/Lab

BLE Exampe Apps

Test App	Results	Test Details
bt_throughput_test	FAIL	Rx and Tx tput measurements. Connection is unstable during traffic with Neon B0 as central.
nimble_beacon	PASS	non connectable advertisement
nimble_connection	PASS	Connectable advertisement and connection
nimble_gatt_server	PASS	Connectable advertisement, connection and GATT operation - read/write/Notify/indicate
nimble_security	PASS	Connectable advertisement, connection and GATT operation - read/write/Notify/indicate; HRS to read/write with security 6-digit pin enabled
nimble_scanner	PASS	Scans for the BLE devices and extract address, RSSI advertisement payload fields — device name, service UUIDs, manufacturer data, etc.
nimble_external_adv_ADV_EXT_MAX_EVENTS	PASS	100 events for extended advertisement
nimble_external_adv_ADV_SCANNABLE_LEGACY_EXT	FAIL	Simple non-connectable scannable instance using legacy PUDs. No advertisements observed.
nimble_external_adv_ADV_LEGACY_DURATION	FAIL	Starts advertising instance with 5sec timeout and changing adv data pattern. No advertisements observed.
nimble_external_adv_ADV_SCANNABLE_EXT	FAIL	This is simple scannable instance that runs forever. No advertisements observed.
nimble_external_adv_ADV_PERIODIC	FAIL	Start periodic advertise packets. No advertisements observed.
ble_dynamic_service	PASS	GATT server, advertisement, add/del custom service tested.

- – DUT: Neon B0 NEB0PP; REF1: Neon B0 NEB0PP
- Test Setup: Cabled inside RF Shield box; Clean channel

BLE Peak Tput

PHY	Packet Length	Connection Interval	Tx Tput (Central) in Kbps	Rx Tput (Peripheral) in Kbps
1M	251	100ms	331	331
2M	251	100ms	1264	1269
LRS2	251	100ms	304	312
LRS8	251	100ms	96	104

- DUT (Peripheral): Neon B0 NEB0PP52
- REF (Central): **Helium A0 HEA0PP59**
- Test Setup: Cabled inside RF Shield box; Clean channel
- RSSI: -33 dBm

BLE Interoperability

Test App	Samsung - Andriod	iPhone15 - iOS	BLE adapter – Linux (Model: TBA)	IFX Devkit (Model: TBA)	Test Details
nimble_beacon	PASS	PASS	PASS	PASS	
nimble_scanner	BLOCKED	BLOCKED	PASS	PASS	Mobile app issue
nimble_connection	PASS	PASS	PASS	FAIL	Connection unstable with IFX devkit and built-in BLE adapter on Linux PC during GATT operations.
nimble_security	PASS	PASS	FAIL	FAIL	
nimble_gatt_server	PASS	PASS	FAIL	FAIL	
nimble_dynamic_service	PASS	PASS	FAIL	FAIL	
nimble_external_adv_ADV_EXT_MAX_EVENTS	PASS	PASS	PASS	PASS	

- DUT: Neon B0 NEB0PP
- Test Environment: OTA testing in noisy RF conditions, BLR office

Peripherals

Example App	Result	Comments
gpio	FAIL	No prints in console to verify app functionality
sysPM	PASS	DUT enters deep power sleep – verified with console prints.
i2c	FAIL	Pre-built app not in SDK. Compilation fails.
task_wdt	FAIL	Pre-built app not in SDK. Compilation fails.
timer_group	FAIL	Pre-built app not in SDK. Compilation fails.

- DUT: Neon B0RK11 EVB

Critical Open Issues

JIRA	Issue	Comments
NSW6-262	Data stalls immediately due to ‘EVENT_WIFI_RX_OVERRUN’ during TCP DL traffic.	Observed on 5/9 APs tested
NSW6-204	‘WIFI_COP_WATCHDOG_TIMEOUT’ observed during TCP UL/DL traffic tests	Observed within a min with Aruba AP534 and TP-Link BE6500
NSW6-174	Disconnects are observed within a min during TCP UL traffic.	Observed on 6/9 APs tested
NSW6-200	UDP DL peak tput is very low with 90M UDP BW	Better with 50M
NSW6-24	RvR sensitivity is low for all traffic types	Around -84 dBm
NSW6-235	DUT fails to connect to AP with WPA2-Personal (PMF Required) / WPA3-SAE securities.	
NSW6-199	BLE connection is unstable during tput and gatt operations with all PHYs	
NSW6-267	Boot argument configuration does not work with Download Tool	

pen issues - https://innophaseiot.atlassian.net/issues/?filter=13347

WiFi Tx Power Tests

TX_POWER_TEST			Neon B0 33# 20260108									Neon B0 29# 20251126									20260108 Avg	20251126 Avg
			test1			test2			test3			test1			test2			test3				
Protocol	Modulation	Design Spec	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462		
802.11b	DSSS-1		16.80	17.05	17.01	16.76	17.00	17.00	16.77	17.00	17.07	16.86	16.89	16.83	16.84	17.02	16.87	16.69	16.88	16.92	16.94	16.87
	DSSS-2		16.79	17.05	17.12	16.76	17.09	17.00	16.85	17.02	16.98	16.71	16.90	16.83	16.75	16.91	16.87	16.65	16.84	16.93	16.96	16.82
	CCK-5.5		15.75	15.97	15.95	15.87	16.08	15.95	15.71	15.95	15.89	15.60	15.80	15.72	15.61	15.83	15.77	15.64	15.94	15.77	15.90	15.74
	CCK-11		15.77	15.98	15.95	15.89	15.98	16.09	15.70	15.96	15.93	15.64	15.83	15.78	15.59	15.84	15.78	15.79	15.83	15.78	15.92	15.76
802.11g	OFDM-6		17.05	17.20	17.21	17.05	17.26	17.23	17.03	17.28	17.26	16.99	17.15	17.06	16.93	17.05	17.11	16.94	17.09	17.08	17.18	17.04
	OFDM-9		16.95	17.22	17.17	16.97	17.15	17.16	17.00	17.24	17.19	16.82	17.06	17.08	16.86	17.02	16.95	16.77	17.03	17.03	17.12	16.96
	OFDM-12		16.93	17.10	17.13	16.89	17.05	17.07	16.73	17.06	17.09	16.79	16.94	16.91	16.77	16.97	16.88	16.72	16.88	16.94	17.01	16.87
	OFDM-18		16.70	16.88	16.90	16.69	16.92	16.96	16.71	16.91	16.92	16.62	16.77	16.70	16.56	16.75	16.75	16.59	16.71	16.69	16.84	16.68
	OFDM-24		16.50	16.77	16.73	16.49	16.76	16.77	16.50	16.74	16.69	16.36	16.51	16.52	16.31	16.63	16.56	16.32	16.60	16.51	16.66	16.48
	OFDM-36		16.32	16.57	16.54	16.29	16.51	16.52	16.28	16.54	16.60	16.27	16.40	16.24	16.27	16.28	16.32	16.20	16.36	16.29	16.46	16.29
	OFDM-48		16.00	16.25	16.30	16.00	16.38	16.24	16.08	16.29	16.22	16.02	16.18	16.12	15.90	16.16	16.17	15.89	16.30	16.11	16.20	16.09
	OFDM-54		15.93	16.12	16.03	15.85	15.94	16.05	15.85	16.02	16.18	15.75	15.86	15.97	15.71	16.00	15.94	15.73	16.02	15.86	16.00	15.87
802.11n	MCS0		15.23	15.49	15.51	15.28	15.48	15.44	15.28	15.43	15.49	15.12	15.40	15.31	15.19	15.43	15.39	15.17	15.35	15.32	15.40	15.30
	MCS1		15.29	15.47	15.47	15.30	15.47	15.51	15.29	15.51	15.46	15.19	15.37	15.26	15.19	15.38	15.33	15.09	15.36	15.31	15.42	15.27
	MCS2		15.24	15.47	15.43	15.30	15.47	15.48	15.29	15.50	15.48	15.13	15.31	15.27	15.18	15.30	15.36	15.12	15.30	15.32	15.41	15.25
	MCS3		15.10	15.40	15.32	15.15	15.33	15.42	15.29	15.31	15.41	15.07	15.24	15.27	15.09	15.27	15.26	15.04	15.27	15.26	15.30	15.20
	MCS4		15.16	15.33	15.32	15.12	15.38	15.38	15.11	15.28	15.28	15.08	15.14	15.13	14.90	15.15	15.12	14.87	15.17	15.13	15.26	15.08
	MCS5		15.02	15.24	15.20	15.03	15.16	15.19	14.97	15.28	15.13	14.89	15.12	15.05	14.91	15.08	15.13	14.83	15.11	15.10	15.14	15.02
	MCS6		14.83	15.03	15.05	14.99	15.04	15.15	14.89	15.14	15.08	14.77	15.02	14.87	14.88	14.96	15.00	14.69	14.90	14.99	15.02	14.90
	MCS7		14.69	14.96	15.01	14.76	14.86	14.99	14.73	14.93	14.99	14.65	14.90	14.87	14.61	14.73	14.84	14.63	14.86	14.81	14.88	14.77
802.11ax	MCS0		17.88	18.18	18.16	17.95	18.15	18.18	17.88	18.13	18.14	17.81	17.95	17.98	17.83	18.00	17.90	17.84	17.91	17.87	18.07	17.90
	MCS1		17.89	18.11	18.12	17.92	18.07	18.14	17.92	18.09	18.03	17.78	18.03	17.91	17.78	17.92	17.86	17.75	17.96	17.97	18.03	17.88
	MCS2		17.79	18.01	17.98	17.77	17.97	18.03	17.78	17.99	17.98	17.69	17.81	17.87	17.68	17.84	17.83	17.64	17.85	17.81	17.92	17.78
	MCS3		17.73	17.92	17.85	17.65	17.87	17.82	17.66	17.82	17.92	17.50	17.71	17.58	17.50	17.71	17.64	17.49	17.68	17.67	17.80	17.61
	MCS4		17.39	17.47	17.53	17.46	17.59	17.63	17.35	17.59	17.59	17.27	17.45	17.42	17.23	17.48	17.45	17.23	17.45	17.40	17.51	17.38
	MCS5		17.10	17.35	17.26	17.23	17.45	17.32	17.12	17.41	17.32	17.09	17.24	17.26	17.00	17.32	17.19	17.05	17.16	17.26	17.29	17.17
	MCS6		16.89	17.08	17.05	16.91	17.40	17.47	16.92	17.15	17.15	16.78	16.90	16.92	16.76	16.85	16.92	16.74	16.99	16.89	17.11	16.86
	MCS7		16.66	16.82	16.78	16.56	16.90	16.86	16.55	16.83	16.83	16.49	16.73	16.68	16.46	16.66	16.70	16.45	16.67	16.62	16.76	16.61
	MCS8		10.97	11.23	11.24	11.11	11.24	11.30	11.13	11.27	11.31	10.98	11.07	11.02	10.89	11.12	11.07	10.90	11.06	11.14	11.20	11.03
	MCS9		11.06	11.33	11.24	11.03	11.31	11.25	11.08	11.30	11.27	11.00	11.08	11.05	10.85	11.15	11.12	10.82	11.10	11.10	11.21	11.03

WiFi Rx PER

RX_PER_TEST			Neon B0 33# 20260108									Neon B0 29# 20251119									20260108 Avg	20251119 Avg
			test1			test2			test3			test1			test2			test3				
Protocol	Modulation	Baseline	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462		
802.11b	DSSS-1	8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	9.33	0.00	1.67	8.33	2.33	3.33	3.00	3.67	0.00	3.74
	DSSS-2	8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	4.33	0.33	0.67	17.67	0.67	5.00	4.00	3.33	0.00	4.11
	CCK-5.5	8	7.33	41.00	1.33	15.33	35.00	2.00	9.67	34.67	4.00	4.67	28.00	1.67	6.00	32.67	2.33	8.33	12.67	12.00	16.70	12.04
	CCK-11	8	0.00	1.00	0.00	0.00	0.67	0.00	0.00	1.67	0.00	3.00	24.00	8.00	3.33	18.00	5.00	5.67	10.33	9.67	0.37	9.67
802.11g	OFDM-6	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	30.67	4.67	1.67	10.33	4.33	5.00	13.33	4.67	0.00	8.41
	OFDM-9	10	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.00	0.00	6.67	86.67	4.67	6.67	24.33	3.33	10.33	14.67	8.33	0.04	18.41
	OFDM-12	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	74.33	5.00	8.67	4.00	3.33	7.67	20.33	5.67	0.00	14.56
	OFDM-18	10	0.00	0.00	0.00	0.00	0.00	0.33	0.00	0.00	0.00	1.33	19.00	2.33	4.00	11.00	2.67	9.33	12.00	16.33	0.04	8.67
	OFDM-24	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.33	11.00	4.67	2.00	3.33	3.33	8.00	14.00	7.33	0.00	6.11
	OFDM-36	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	13.33	3.00	1.67	12.00	4.33	6.67	49.33	0.00	0.00	10.11
	OFDM-48	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.67	5.33	7.33	0.00	1.33	8.67	19.67	2.33	0.00	7.48
	OFDM-54	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.33	15.33	0.00	2.67	1.67	6.00	26.33	13.00	2.00	0.00	7.59
802.11n	MCS0	10	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.20	2.00	11.20	3.80	3.80	11.40	2.20	8.40	13.20	8.40	0.04	7.16
	MCS1	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.20	0.40	9.80	4.20	5.20	7.00	3.40	6.40	10.00	9.80	0.02	6.24
	MCS2	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.20	0.60	9.00	2.60	0.80	0.60	5.20	0.80	23.60	5.60	0.02	5.42
	MCS3	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.80	26.40	4.00	1.40	3.60	4.00	6.40	13.20	4.60	0.00	7.16
	MCS4	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.80	5.00	2.40	1.40	4.40	2.60	4.60	16.60	5.40	0.00	4.80
	MCS5	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.40	9.00	4.00	1.40	9.60	3.20	9.00	17.80	3.20	0.00	6.40
	MCS6	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60	14.00	3.40	22.20	11.20	3.40	0.60	11.20	3.40	0.00	7.78
	MCS7	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.40	29.20	4.00	0.80	9.20	3.80	6.20	8.20	5.20	0.00	7.78
802.11ax	MCS0	10	0.20	0.40	1.00	0.60	1.80	1.40	0.60	1.20	0.80	2.20	20.80	5.40	11.40	10.20	5.40	17.20	25.00	4.60	0.89	11.36
	MCS1	10	2.40	2.20	3.20	1.80	1.40	3.60	2.40	0.80	2.80	4.40	32.00	6.60	9.40	8.60	5.40	9.00	24.40	4.60	2.29	11.60
	MCS2	10	0.60	0.60	0.60	0.40	0.60	0.60	0.60	0.80	0.60	2.00	35.40	7.00	7.00	17.40	4.20	7.60	10.60	4.80	0.60	10.67
	MCS3	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.80	9.80	3.40	1.00	7.40	2.40	4.80	9.80	5.20	0.00	5.18
	MCS4	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60	12.80	4.60	0.80	10.20	3.00	5.60	25.00	4.00	0.00	7.40
	MCS5	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.80	67.20	0.00	0.60	10.80	3.80	7.00	15.00	5.40	0.00	12.29
	MCS6	10	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.80	17.40	0.20	0.80	13.00	4.00	7.20	25.00	3.40	0.02	7.98
	MCS7	10	0.00	5.00	0.00	0.00	2.00	0.00	0.00	0.80	0.00	1.60	4.40	0.40	1.60	5.80	2.60	5.00	14.00	2.20	0.87	4.18
	MCS8	10	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60	21.00	4.60	1.20	10.40	4.80	13.00	20.60	2.60	0.02	8.76
	MCS9	10	0.00	1.20	0.40	0.00	1.40	0.40	0.00	1.40	0.20	0.60	14.00	0.80	3.80	10.60	2.40	8.60	11.80	5.00	0.56	6.40

WiFi Rx Sensitivity

RX_SENSITIVE_TEST			Neon B0 33# 20260108									Neon B0 29# 20251119									20260108 Avg	20251119 Avg
			test1			test2			test3			test1			test2			test3				
Protocol	Modulation	Baseline	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462	2412	2437	2462		
802.11b	DSSS-1	-80	-91	-91	-91	-91	-91	-91	-91	-91	-91	-77	-78	-91	-89	-83	-88	-77	-82	-82	-91	-83
	DSSS-2	-80	-91	-90	-91	-91	-91	-91	-91	-91	-91	-87	-83	-85	-89	-82	-84	-81	-77	-79	-91	-83
	CCK-5.5	-76	-75	NULL	-86	-75	NULL	-86	-75	NULL	-86	-78	NULL	-82	-75	NULL	-84	NULL	NULL	NULL	-81	-80
	CCK-11	-76	-84	-78	-84	-84	-78	-84	-84	-79	-84	-79	NULL	-83	-78	NULL	-80	NULL	NULL	-73	-82	-79
802.11g	OFDM-6	-82	-91	-88	-91	-91	-88	-91	-91	-88	-91	-86	-79	-88	-86	-81	-80	-81	NULL	-80	-90	-83
	OFDM-9	-81	-89	-86	-88	-89	-86	-89	-89	-86	-88	-83	-79	-88	-84	-79	-85	-82	NULL	-86	-88	-83
	OFDM-12	-79	-88	-85	-87	-88	-85	-88	-88	-85	-87	-86	-77	-78	-84	-81	-77	-77	-77	-79	-87	-80
	OFDM-18	-77	-86	-83	-85	-86	-83	-85	-86	-83	-85	-85	-76	-76	-82	-77	-85	-80	NULL	-76	-85	-80
	OFDM-24	-74	-80	-80	-80	-80	-80	-80	-80	-80	-80	-77	-74	-75	-79	-70	-80	-73	-70	-80	-80	-75
	OFDM-36	-70	-76	-76	-76	-76	-76	-76	-76	-76	-76	-76	-66	-76	-74	-74	-76	-70	-67	-75	-76	-73
	OFDM-48	-66	-70	-70	-70	-70	-70	-70	-70	-70	-70	-70	-63	-70	-70	-65	-70	-70	-62	-70	-70	-68
	OFDM-54	-65	-70	-70	-70	-70	-70	-70	-70	-70	-70	-70	-61	-68	-70	-67	-70	-64	-60	-61	-70	-66
802.11n	MCS0	-82	-90	-87	-90	-90	-86	-90	-90	-87	-90	-88	-80	-82	-84	-82	-89	-83	NULL	-86	-89	-84
	MCS1	-79	-87	-85	-87	-87	-85	-87	-87	-85	-87	-87	-83	-85	-81	-82	-82	-82	-77	-85	-86	-83
	MCS2	-77	-85	-82	-85	-85	-82	-85	-85	-82	-85	-84	-75	-81	-81	-77	-85	-80	-79	-77	-84	-80
	MCS3	-74	-81	-79	-81	-81	-79	-81	-81	-79	-81	-81	-73	-80	-79	-76	-81	-77	-71	-79	-80	-77
	MCS4	-70	-77	-75	-77	-77	-76	-77	-77	-75	-77	-77	-66	-77	-75	-72	-77	-73	-65	-76	-76	-73
	MCS5	-66	-73	-71	-73	-73	-72	-73	-73	-72	-73	-73	-62	-73	-71	-68	-73	-63	-63	-72	-73	-69
	MCS6	-65	-71	-70	-71	-71	-70	-71	-71	-70	-71	-71	-61	-71	-68	-65	-70	-67	-61	-67	-71	-67
	MCS7	-64	-71	-68	-71	-71	-68	-71	-71	-68	-71	-71	-59	-71	-67	-59	-71	-65	-60	-59	-70	-65
802.11ax	MCS0	-82	-86	-86	-86	-87	-86	-86	-87	-86	-86	-84	-77	-85	-83	-81	-85	-84	-78	-77	-86	-82
	MCS1	-79	-83	-82	-82	-83	-82	-82	-83	-81	-82	-81	-74	-78	-80	-78	-82	-77	-76	-80	-82	-78
	MCS2	-77	-83	-78	-82	-83	-78	-83	-83	-78	-83	-80	-72	-79	-75	-75	-72	-72	NULL	-72	-81	-75
	MCS3	-74	-81	-77	-81	-81	-77	-81	-81	-78	-80	-78	-69	-74	-75	-72	-78	-74	-68	-69	-80	-73
	MCS4	-70	-77	-72	-77	-77	-72	-77	-77	-72	-77	-74	-67	-77	-72	-69	-77	-72	-69	-69	-75	-72
	MCS5	-66	-73	-69	-73	-73	-69	-73	-73	-69	-73	-72	NULL	-73	-70	-65	-73	-62	-63	-61	-72	-67
	MCS6	-65	-71	-67	-71	-71	-67	-71	-71	-67	-71	-71	-63	-71	-69	-63	-70	-66	NULL	-63	-70	-67
	MCS7	-64	-71	-64	-70	-70	-65	-70	-70	-65	-70	-69	-61	-64	-64	-61	-70	-63	-59	-68	-68	-64
	MCS8	-59	-65	-61	-65	-66	-62	-64	-66	-62	-65	-64	-54	-55	-62	-56	-64	-64	-54	-64	-64	-60
	MCS9	-57	-62	-58	-62	-63	-58	-62	-63	-59	-62	-58	-53	-59	-59	-56	-62	-57	-50	-59	-61	-57

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